Geopolitical Distance and High Seas Fishing*

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November 8, 2026

Abstract

To what extent is resource use in transboundary commons driven not only by economic incentives but also by geopolitical ones? We extend the Gordon-Schaefer fisheries model to include political payoffs and show that these can sustain resource extraction even under negative economic returns. We test the framework on the Chinese distant-water fishing fleet, the world's largest, by examining its activity on the high seas just outside foreign exclusive economic zones. Holding economic incentives constant, the probability of Chinese fleet presence or fishing rises with the UN General Assembly voting-based political distance between China and the coastal host country. The pattern is consistent with surveillance as the main motive: the effect is concentrated in small fleets of vessels with untraceable ownership that, rather than prioritizing economic incentives, have a tendency to cluster near submarine cables and NATO military installations.

JEL codes: F51, F52, Q22, Q34, H56
Keywords: distant-water fishing, high seas, global commons, geopolitics, economic statecraft, political distance, surveillance, China

*We thank Matthias Flückiger, Martin Gassebner, Patrick Hufschmidt, Lennart Kaplan, Christopher Kilby, Christian Leßmann, Markus Ludwig, Jingke Pan, Vice Admiral Robert L. Thomas, Felix Turbanisch, audiences at VfS Conference 2025, Kiel-Göttingen-CEPR Conference on China in the Global Economy 2025, Bundesbank Workshop for Young Scientists on Applied Economics 2025, Applied Young Economists Webinar 2025, Political Economy of Democracy and Dictatorship Conference 2024, European Public Choice Society Meeting 2024, and many seminars for constructive feedback. All mistakes remain ours.

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1 Introduction

Distant-water fishing (DWF) on the high seas, which covers about two-thirds of the global ocean, has increased markedly over recent decades. From 1950 to 2014, major fishing fleets expanded their average traveling distance by up to 4,000 km (Tickler et al., 2018). These fleets depend heavily on government subsidies, without which they would often operate at a loss (Sala et al., 2018). Such extensive state support for extraction in a weakly governed commons raises a question: is resource use in the global commons driven solely by economic incentives, or also by geopolitical ones? This question speaks to a central assumption of existing commons-governance frameworks. Economic models, from the open-access problem to its proposed solutions such as cooperative management of shared stocks or property rights, treat extracting agents mainly as seeking to maximize resource rents (Gordon, 1954; Munro, 1979; Costello et al., 2008). By contrast, the literature on economic statecraft, and the growing work on geoeconomics in particular, shows that states use commercial activity to advance geopolitical objectives, largely by directly controlling access to something the target country depends on (Baldwin, 1985; Farrell and Newman, 2019; Mohr and Trebesch, 2025).[1] State-supported DWF fleets may serve such objectives in a different way. Because controlling access is not possible on the high seas, fleets can instead generate strategic benefits through sustained presence and extraction. If geopolitical incentives sustain fleet activity beyond what is economically viable, conventional commons models may understate pressure on marine resources and overstate the prospects for cooperative governance.

This paper examines whether geopolitical divergence between countries shapes distant-water fishing in the global commons of the high seas beyond economic incentives. To answer this question empirically, we study deployment of the Chinese DWF fleet near foreign exclusive economic zones (EEZs). China operates the world's largest DWF fleet and one of the most heavily subsidized (Sala et al., 2018; Millage et al., 2021). Although the fleet is estimated to operate at a loss, sustained government support is justified on grounds of food security, employment, seafood exports, and broader strategic objectives including the "going out" policy and "Marine Power Strategy" (Mallory, 2013, 2016; Englander et al., 2025). Existing anecdotal accounts suggest that the role of the fleet extends beyond commercial objectives. It has been described as an extension of China's maritime security apparatus and foreign policy (Office of Naval Intelligence, 2021; Macias and Jensen, 2025), particularly through the

[1] Documented instruments primarily exploit control over markets, resources, or infrastructure on which target countries depend, including sanctions (Felbermayr et al., 2021), tariffs (Fetzer and Schwarz, 2021), import dependence and export restrictions (Fan et al., 2023; Liu and Yang, 2025), network and digital chokepoints (Farrell and Newman, 2019; Porcellacchia et al., 2026), financial chokepoints (Matvos and Neiman, 2026), and infrastructure investment (Dreher et al., 2022).

maritime militia—nominally commercial vessels with navy-trained crew members that can be mobilized for state security. Yet systematic, quantitative evidence on whether geopolitical considerations shape DWF deployment, and on the nature of those incentives, remains scarce.

We consider four political channels through which DWF deployment may be motivated. First is enforcement, in which the risk of coercion by the host state may deter the foreign fleet's activity near the host's waters. In contrast, fleet activity in these areas may generate political benefits for the vessel-sending state. The second and third channels hence capture potential political gains from power projection and surveillance. These mirror an ex-ante defense logic in which, as relationships worsen, the sending state has greater incentives to signal operational capability to intimidate and deter (Fearon, 1997) or to increase monitoring (Hoffman, 2002). The fourth channel is retaliation, whereby diplomatic transgressions are punished in a tit-for-tat fashion by increased vessel presence that pressures local fishing stocks (Axelrod, 1980). We incorporate these political costs and benefits into the Gordon-Schaefer fisheries model as additional determinants of optimal fishing effort and show that net political gains can sustain resource extraction even when economic returns are negative.

Our empirical analysis focuses on Chinese DWF activity on the high seas just outside EEZs where Chinese vessels have no fishing access. In these areas, Chinese activity is close enough to matter strategically to the EEZ host state, yet the host has no legal means to prevent it. We exclude EEZs that grant Chinese vessels access and those with overlapping claims, as activity just outside the former may reflect spillover from licensed fishing instead of strategic positioning and the latter have no clearly identifiable host state. Using Global Fishing Watch data, we observe Chinese fishing vessels from 2012 to 2020 in 50 km buffers outside 71 EEZs of 21 countries. Our identification exploits variation over time in political distance between China and the host of a given EEZ, defined to be the absolute difference between their UN General Assembly (UNGA)-based ideal points (Bailey et al., 2017). Conditional on host country and EEZ time-invariant characteristics, common shocks, and the economic incentives to fish proxied by concurrent non-Chinese fishing effort in the buffer and lagged phytoplankton abundance, we compare Chinese activity outside the same EEZ as the host's alignment with China shifts.

We find that a one-point (roughly one SD) larger in China-host political distance two years earlier is associated with a 15 percentage-point increase in the probability of Chinese DWF presence, about 40% of the sample mean.[2] We document a similar pattern for the outcome indicating any active fishing. Our identifying assumption is that, conditional on the included

[2]The two-year lag accounts for the timing of UNGA voting sessions and allows political dynamics to translate into changes in fleet activity. We provide a detailed discussion in Section 5.

controls, variation in China-host political distance is uncorrelated with unobserved determinants of Chinese DWF activity just outside the EEZ. This could be violated by reverse causality, omitted confounders, or measurement error. A distributed-lag specification first rules out reverse causality: only the two year-lag of political distance returns a statistically significant positive coefficient, while leads and contemporaneous measure do not. Second, the estimates are stable when we add ecological and bilateral economic controls. Political distance also does not predict the activity of the comparable Taiwanese and Japanese fleets, indicating a China-specific relationship rather than general economic or regulatory drivers of DWF. Third, alternative measures of political distance yield similar results, which supports that our estimates reflect actual political divergence. Finally, we implement a two-stage least squares strategy and instrument political distance with China-host disagreement on UNGA nuclear-weapons votes (Voeten, 2013). This yields a significant first stage and an estimate consistent with the baseline, providing evidence that the relationship is not driven by economic factors that shape both UN voting and fleet activity. A broad set of robustness checks further show that our results are not driven by outliers or data-construction choices.

At the intensive margin, we find that the number of Chinese vessels in a buffer-month is significantly and positively associated with political distance. Estimates for presence and fishing hours are imprecise in the raw data but become positive and statistically significant after winsorizing these highly skewed outcomes at the 99th or 95th percentile. Similar patterns emerge when we exclude host countries with the highest recorded hours. These countries have exceptionally productive fishing grounds or longstanding bilateral fisheries agreements with China, making fleet activity around their EEZs plausibly less responsive to changes in political distance. Combined, these findings suggest that widening political distance between China and a host country is associated with both a greater likelihood and a greater intensity of Chinese DWF deployment immediately outside the host's EEZ.

To inspect the political motivations underlying these results, we evaluate four candidate mechanisms: enforcement, power projection, retaliation, and surveillance. The evidence most consistently supports the surveillance channel. Enforcement would predict Chinese vessels to visit borders less when relationships worsen due to increased scrutiny from the host country, which is the opposite of what we find. This indicates that, at best, enforcement effects are outweighed by political payoffs. Fleet-level heterogeneity further rules out power projection and retaliation as the main political motives. The effects are concentrated in small, inconspicuous fleets operating near democratic states, rather than in the large, visible, and economically disruptive deployments one would expect if power projection or retaliation were the primary objective. Additionally, using a severity index built from GDELT event data,

we find that China responds to diplomatic transgressions not by retaliating but by taking a conciliatory stance and reducing large fleet deployments. Crucially, we collect rich vessel-level data and show that vessels that are the most sensitive to political incentives stand out along several dimensions compared to the remaining fleet. These vessels tend to lack traceable ownership and are disproportionately equipped with trawler gear, consistent with purpose-built Maritime Militia platforms (Poling et al., 2021). Moreover, instead of following economic incentives, they spend significantly more time near submarine cables and NATO military installations. Together, these findings point to a subset of operationally focused vessels plausibly gathering intelligence on strategic infrastructure and military assets as the primary driver of the positive association between political distance and fleet activity.

Contributions. Our paper combines two literatures that have developed independently. Research on common-pool resources explains when shared stocks are overexploited and when cooperation can be sustained, but generally only considers economic returns as incentives (Gordon, 1954; Schaefer, 1954, 1957; Hardin, 1968; Munro, 1979; Ostrom, 1990; Costello et al., 2008; Stavins, 2011). Economic statecraft, as part of the broader geoeconomics literature, explores how states pursue geopolitical aims through economic means and mostly examines instruments that operate through jurisdiction over scarce resources and networks the target depends on (Baldwin, 1985; Farrell and Newman, 2019; Felbermayr et al., 2021; Fetzer and Schwarz, 2021; Liu and Yang, 2025; Mohr and Trebesch, 2025; Porcellacchia et al., 2026). Neither alone fully accounts for the incentives underlying DWF activity. By extending the canonical Gordon-Schaefer model to incorporate geopolitical payoffs, we show that these can sustain extraction even where economic returns are negative, thus accelerating stock degradation. These political determinants of fishing behavior have direct implications for fishery governance. We further contribute to the geoeconomics literature by identifying a novel instrument suited to the global commons. Specifically, our evidence indicates that fishing vessels on the high seas can advance geopolitical objectives through intelligence gathering and surveillance. While espionage is not new to the geoeconomics literature, previous work has focused on industrial espionage (Glitz and Meyersson, 2020; Mohr and Trebesch, 2025). The instrument we identify differs substantially in both use case and utility.

The paper also speaks to the emerging literature on the economic and political drivers of high-seas fishing (Zhang, 2016; Sala et al., 2018; Schuhbauer et al., 2020; Englander et al., 2025), and to research on interstate relations, political preference divergence, and conflict (Gartzke, 1998; Oneal and Russett, 1999; Sweeney, 2003; Kastner, 2007; Dreher et al., 2008; Kilby, 2009; Dreher et al., 2009, 2018), where we identify vessel deployment in international waters as an understudied channel of geopolitical competition.

The paper continues as follows. Section 2 provides the institutional background. Section 3 develops a conceptual framework for the possible political incentives of DWF activity. Section 4 describes the data sources and variable construction, while Section 5 outlines the empirical framework. Sections 6 and 7 present the main results and explore the underlying mechanisms, respectively. Section 8 discusses the policy implications, and Section 9 concludes.

2 Background

Distant water fishing. Distant water fishing (DWF) refers to fishing conducted outside a country's jurisdictional waters, either on the high seas or within other countries' exclusive economic zones (EEZs).[3] As coastal stocks have declined in many regions, fishing fleets have moved farther offshore and onto the high seas (Norse et al., 2012). Between 1950 and 2014, major industrial fishing states such as Taiwan, South Korea, Spain, and China expanded their average distances to fishing grounds by 2,000–4,000 km (Tickler et al., 2018). This outward expansion has raised concerns about environmental degradation, resource depletion, and weak governance (McWhinnie, 2009; Norse et al., 2012), as well as socio-economic losses for coastal communities and broader risks to security and political stability (Bennett et al., 2018; Sumaila et al., 2015; Bailey, 1997; Sanjeet, 2013; Ewell et al., 2017).

China operates the world's largest DWF fleet with 4,615 vessels, followed by the European Union (964) and Taiwan (634) (Millage et al., 2021). China's DWF industry emerged in 1985, after key domestic commercial stocks declined during the 1970s and 1980s (Mallory, 2013, 2016), and has seen substantial expansion over time (see Appendix Figure B.1). The literature points to three main motives for its expansion. First, the fleet serves food security: with overfished domestic waters and rising seafood demand, China's EEZ alone can no longer meet consumption needs (Mallory, 2013, 2016). Second, it sustains the domestic fishing industry, employment, and seafood exports (Mallory, 2013). Third, DWF aligns with broader state objectives, including the "going out" policy encouraging Chinese firms to expand abroad and the "Marine Power Strategy" aimed at establishing China as a major maritime power (Mallory, 2016; Englander et al., 2025).

DWF fleets depend crucially on government subsidies. Sumaila et al. (2016) estimate global fisheries subsidies at around 35 billion USD (2009 prices), with fuel taking up the largest

[3]The high seas are the internationally shared marine areas beyond national jurisdictions and account for around 65% of the global ocean surface (Sala et al., 2018). By contrast, exclusive economic zones (EEZs), established under the United Nations Convention on the Law of the Sea, extend up to 200 nautical miles from the coastlines of states and their islands. Within EEZs, coastal states hold exclusive rights over the protection and management of marine resources, whereas the high seas remain largely unregulated and are accessible to vessels from all countries.

part, followed by management and port building. For DWF on the high seas, China provides the largest subsidy volume worldwide—around 1.7 billion USD (2018 prices)—followed by Japan at 695 million USD (Millage et al., 2021). The ecological consequences and economic inefficiency of these subsidies are well documented (Sumaila et al., 2016; Sala et al., 2018; Costello et al., 2021; Englander et al., 2025). Sala et al. (2018) estimate that 2014 global high-seas fishing profits ranged from 3.8 to 5.6 billion USD with subsidies, but from -364 million to 1.4 billion USD without. The pattern is starker for China: 2014 high-seas profits ranged from 1 to 479 million USD with subsidies and from -418 to 60 million USD when they are netted out (Sala et al., 2018). Despite their limited market profitability, why do states support DWF fleets? Explanations include potential under-reported catches that mask actual profits, labor and regulatory cost advantages, and seasonal markups (Sala et al., 2018), alongside justifications grounded in food security, employment, and strategic considerations (Mallory, 2013, 2016; Englander et al., 2025).

Maritime militia. Building on the economic and developmental purposes of DWF discussed earlier, a growing body of work argues that Chinese fishing vessels may also serve foreign policy and security functions. Central to this debate is China's Maritime Militia, a force of civilians who receive military training from the People's Liberation Army (PLA) Navy and the Chinese Coast Guard and who can be mobilized for maritime missions (Luo and Panter, 2021; Martinson, 2025). By relying on civilian fishing vessels, the militia engages in so-called gray-zone operations that advance security objectives while reducing the risk of overt military escalation (Luo and Panter, 2021). Importantly, these vessels do not cease to be commercially motivated, but continue to retain economic incentives to fish alongside their security-related functions (Luo and Panter, 2021). Despite their poor combat ability, these units are strategically relevant because they can support reconnaissance and the safeguarding of China's maritime claims well beyond China's immediate coastline (Luo and Panter, 2021). Fishing vessels are particularly useful in this regard because they can maintain a persistent presence in strategically relevant waters while operating under civilian cover and plausibly justifying that presence on economic grounds (Protzer and Stein, 2023; Martinson, 2025).

The use of fishing vessels in this way has led some sources to describe parts of China's maritime fishing presence as an extension of its foreign policy and maritime security apparatus (Office of Naval Intelligence, 2021; Macias and Jensen, 2025). More recent work from the US Naval War College's China Maritime Report details the extent and structure of the militia's intelligence-gathering functions (Martinson, 2025). Specialized maritime reconnaissance units conduct intelligence, surveillance, and reconnaissance missions under military direction, from reporting unusual activity to close-in reconnaissance of foreign ships. These

operations remain concentrated in China's near seas, but PLA officers have called for integrating distant-water fishing vessels into the militia's intelligence reporting network. Militia intelligence personnel already embark on some DWF vessels (Poling et al., 2021), although systematic public evidence remains scant (Martinson, 2025).

Submarine cables, which carry almost all intercontinental internet traffic, are of special concern and, similar to previous telecommunications infrastructure, interference with them has long been a site of geoeconomic competition (Porcellacchia et al., 2026). Despite their critical intelligence value, tapping cables in the high seas for the purpose of eavesdropping is not prohibited under current international law (Burdette, 2021). Moreover, bottom-gear fishing is believed to be the leading cause of submarine cable faults (Drew and Hopper, 2009).[4] In addition, the aforementioned gray-zone accusations have been made against Chinese-linked vessels over cable breaks that could pass as anchoring or fishing accidents (Braw, 2023; Hale, 2025). Access to cables is of special importance for China: roughly 70% of global data flows today traverse at least one US-owned cable (Porcellacchia et al., 2026). Farrell and Newman (2019) distinguish two forms of leverage conferred by jurisdiction over network hubs: a chokepoint effect, i.e., the ability to deny access, and a panopticon effect, i.e., the ability to observe what passes through. Western states obtain both through jurisdiction over cable operating firms. We hypothesize that, for a state lacking that jurisdiction, physical presence at the cable could substitute for both effects: severance for the chokepoint and fiber tapping for the panopticon.

Anecdotal accounts further describe Chinese militia and fishing vessels as carrying military-grade equipment and as having rammed, used water cannon against, or maneuvered dangerously toward foreign ships (Goodman, 2022; Davidson, 2024). Combined with the sudden appearance of large numbers of fishing vessels near the maritime boundaries of other states, China's DWF presence has been documented to jeopardize the livelihoods of local fishermen while also causing ecological damage (Collyns, 2020; Dubé, 2024; Macias and Jensen, 2025). Under the guise of DWF, the Maritime Militia could therefore plausibly be employed to intimidate foreign vessels, project Chinese power into strategically relevant maritime areas, or respond to diplomatic tensions without overt military escalation.

[4] Cables are especially vulnerable deep waters as they are less likely to be buried or armored there (Drew and Hopper, 2009).

3 Conceptual framework

3.1 Possible political motivations

In light of the background in Section 2, we derive our hypotheses from a game-theory perspective. The starting point is that fishing vessels are civilian actors with economic motives, but can additionally serve as a geoeconomic instrument by generating strategic value. In particular, the same vessel can produce information, demonstrate presence, or impose costs on other states while keeping fishing incentives and operating under a commercial guise. We consider enforcement by EEZ host states, as well as power projection, surveillance, and retaliation on the Chinese side as potential political motivations. Changing bilateral relations between China and the relevant coastal state are leveraged as the main explanatory variable shaping political incentives. If political motivations matter, different mechanisms should yield different empirical predictions for vessel behavior as bilateral relations between China and host states deteriorate.[5] These are not mutually exclusive, and any observed empirical pattern may be explained by a combination of the channels discussed below. Section 7 empirically assesses the plausibility of these mechanisms.

Enforcement. One possible mechanism is enforcement. Here, rather than China, the relevant agency lies with the EEZ host states, who may reduce leniency and enforce maritime rules more aggressively against Chinese vessels. There are numerous examples of this, with measures ranging from increased monitoring to detention and even the sinking of Chinese fishing vessels (Costa and Blanchard, 2016; Reuters, 2016, 2017). In this case, the expected costs of operation rise because enforcement increases both the economic and safety risks faced by fishing vessels. If this mechanism dominates, one would expect both vessel presence and fishing effort to decline as political relations deteriorate. The enforcement channel therefore predicts the opposite empirical pattern from the China-driven channels discussed below.

Power projection. A second possible mechanism is the use of fishing vessels for global power projection. This possibility stems from China's broader strategic ambition to operate as a global maritime power (Mallory, 2016; Englander et al., 2025). Under this logic, Chinese vessel activity serves not only commercial purposes but also the broader political objective of demonstrating China's capacity to maintain a persistent presence in strategically relevant maritime areas. Incentives for power projection should be higher for states with which China has tense or adversarial relations. The value of operating in these waters lies in signaling reach, resolve, and operational capability to coastal states. A striking illustration is the

[5] Worsening relations are used to illustrate our mechanisms. Apart from the retaliation channel, we expect effects to be symmetric when relations improve.

recent formation of a sea barrier more than 300 km long by Chinese fishing vessels in the East China Sea (Buckley et al., 2026). In a more coercive form, large-scale deployments of fishing vessels have been described as overwhelming smaller host countries (Collyns, 2020; Dubé, 2024), thus plausibly becoming instruments of intimidation. The signal value of the power-projecting operations depends on the deployment scale and visibility. Therefore, power projecting vessels are expected to travel in larger fleets that overtly demonstrate presence near borders through retaining their AIS signals. Fishing effort, by contrast, is less likely to rise. Although maritime militia vessels retain economic incentives, active fishing would blur the intent of the deployment and thus weaken the visibility that power projection relies on.

Surveillance. Another possible mechanism is reconnaissance or surveillance. In repeated strategic interactions, trust and verification can act as substitutes. When trust declines, actors have stronger incentives to monitor others (Hoffman, 2002). Applied to the maritime domain, deteriorating interstate relations lower trust, thus increasing the value of monitoring the behavior, capabilities, and intentions of other states. Fishing vessels offer a potentially attractive tool for such monitoring because they can maintain persistent presence in strategically relevant maritime spaces while retaining a civilian appearance. Their commercial guise is especially useful because overt surveillance can itself signal distrust (Falk and Kosfeld, 2006), and a target who knows it is being observed will adopt strategies that conceal the very information the observer seeks (Avenhaus et al., 2002). As detailed above, the qualitative maritime militia literature reinforces the plausibility of this mechanism by emphasizing the reconnaissance and intelligence-gathering functions of Chinese fishing vessels (Luo and Panter, 2021; Protzer and Stein, 2023; Martinson, 2025). What distinguishes the surveillance channel from the other strategic motivations is that worsening bilateral relations should increase the likelihood of small-scale vessel deployments, rather than the appearance of large fleets. Fishing effort may also rise to the extent that it helps legitimize and disguise surveillance missions. We lastly also expect surveillance vessels to spend a disproportionate time near borders or strategic locations, where intelligence collection yields the highest returns, e.g., through movement monitoring, mapping vulnerabilities, and intercepting communications.

Retaliation. The last mechanism examined is retaliation. In repeated games, reciprocal responses such as tit-for-tat can emerge as effective strategies in the face of hostile actions (Axelrod, 1980). For retaliation to be effective, the response must be observable, reasonably prompt, and attributable to the retaliating party, so that the target updates its behavior in the future. In the maritime setting, political disputes may induce China to use its fishing fleet in ways that impose economic costs on rival states. One possible channel is intensified

fishing activity close to foreign EEZs, thereby reducing fish stocks available to local fleets, increasing congestion, or putting pressure on coastal ecosystems and livelihoods. Under this mechanism, worsening interstate relations should be associated with higher vessel presence as well as greater fishing effort. The major distinguishing factor for retaliation is the time-frame of the response and the nature of the interstate relations shifts. Tit-for-tat requires quick and observable responses that the opposing party can causally link to an individual action that the retaliating party wants to disincentivize. Thus, the response should be within months of a politically salient dispute. By contrast, surveillance and power projection are more likely to arise in response to broader and more gradual geopolitical divergence. Retaliation is likely to involve large fleets that are able to inflict meaningful economic damages.

3.2 Modeling political motivations

To formally assess how political motivations could affect the optimal level of DWF fleet's effort, we incorporate the potential political constraints or payoffs in the Gordon-Schaefer bioeconomic model (Gordon, 1954; Schaefer, 1954, 1957). We start with the benchmark model where DWF vessel i chooses effort E_i to maximize its profit. Let $V_i(.)$ denote the vessel's objective function and $\Pi_i(E_i, X)$ its profit, with X being the available fish stock.[6] In the purely economic benchmark,

$$V_i(E_i, X) = \Pi_i(E_i, X) = \underbrace{p_i\, q_i\, X\, E_i}_{\text{total revenue}} - \underbrace{(c_i\, E_i + 0.5\, \gamma_i\, E_i^2)}_{\text{total cost}}. \tag{1}$$

q_i is the catchability coefficient that determines total harvest and p_i is the unit price of harvest. c_i is the baseline marginal cost, and $\gamma_i > 0$ captures increasing marginal costs from travel distance, congestion, and operational complexity. The quadratic cost term deviates from standard Gordon-Schaefer linear cost function to ensure an interior optimum when X is treated as given. As the vessel chooses effort to maximize profit, solving the first-order condition yields the optimal effort level:

$$E_i^* = \frac{p_i q_i X - c_i}{\gamma_i}. \tag{2}$$

[6] In this setup, each individual vessel takes X as parametric, consistent with the framework of Gordon (1954). A single vessel's harvest does not change aggregate stock meaningfully, and over the monthly horizon of our empirical analysis stock dynamics are slow relative to within-month variation in effort. We thus do not close the model with the stock-dynamic equation and simply derive optimal-effort change under additional political motivations while treating X as given.

When effort generates not only economic but political payoffs, the vessel's objective becomes:

$$V_i(E_i, X) = \Pi_i(E_i, X) + \Phi_i(E_i), \tag{3}$$

in which the political payoff is defined as $\Phi_i(E_i) = \alpha_i^{psr} E_i - \alpha_i^e E_i$, with $\alpha_i^{psr}, \alpha_i^e \geq 0$. Consistent with the potential political motives discussed in Section 3.1, α_i^e captures any deterrence induced by enforcement from the host country, while α_i^{psr} captures potential power projection, surveillance, or retaliation motives as gains for the vessel-sending country.[7] The net political payoff is $\Delta_i \equiv \alpha_i^{psr} - \alpha_i^e$. As the vessel maximizes its aggregate objective, we solve the F.O.C and the optimal level of effort under the augmented scenario becomes:

$$E_i^{**} = \frac{p_i q_i X - c_i + \Delta_i}{\gamma_i}. \tag{4}$$

This yields three empirically relevant scenarios to test how optimal level of vessel's effort changes when political payoffs enter the objective function.

1. Neutral political incentives. If $\alpha_i^{psr} = \alpha_i^e$, then $\Delta_i = 0$ and $E_i^{**} = E_i^*$. This implies that political incentives cancel out, and effort is determined only by economic returns.

2. Negative net political incentives. If $\alpha_i^{psr} < \alpha_i^e$, then $\Delta_i < 0$. Political deterrence from host country lowers the marginal payoff from the vessel's effort and the optimal effort level is lower than in the purely economic benchmark: $E_i^{**} < E_i^*$. In this case, fishing may cease even when the economic marginal return is still positive, as long as the total marginal payoff is non-positive:

$$\frac{\partial \Pi_i}{\partial E_i} > 0 \quad \text{and} \quad \frac{\partial \Pi_i}{\partial E_i} + \Delta_i \leq 0. \tag{5}$$

Thus, enforcement can deter fishing before the vessel reaches its profit-maximizing capacity.

3. Positive net political incentives. If $\alpha_i^{psr} > \alpha_i^e$, then $\Delta_i > 0$. Political motives increase the marginal payoff from effort and optimal effort is higher than in the purely economic benchmark: $E_i^{**} > E_i^*$. In this case, fishing may continue even when the economic marginal return is negative, as long as the total marginal payoff remains positive:

$$\frac{\partial \Pi_i}{\partial E_i} < 0 \quad \text{and} \quad \frac{\partial \Pi_i}{\partial E_i} + \Delta_i > 0. \tag{6}$$

Thus, politically motivated effort can persist even under economic losses. Appendix A provides detailed model derivations. In the next sections, we test these hypotheses empirically.

[7] We assume Φ is linear in effort for tractability. This allows the empirical analysis to focus on the sign and heterogeneity of the net political payoff rather than its functional form.

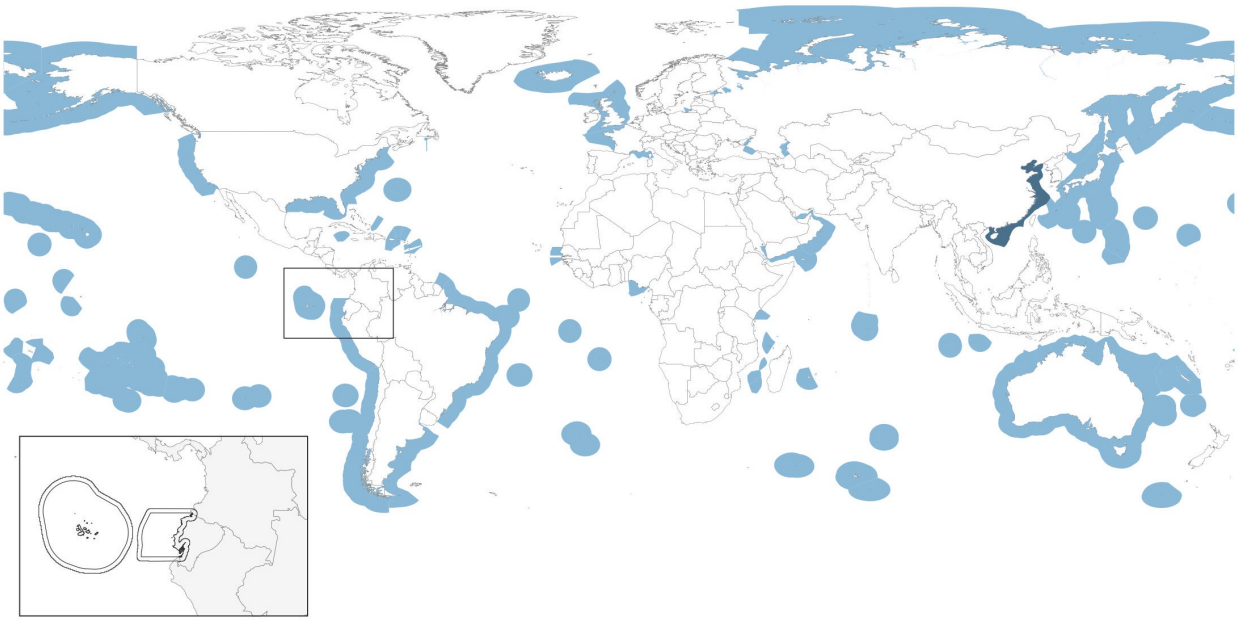


Figure 1: Study area
Figure shows the exclusive economic zones in our sample (lighter shading) and, for reference, China's exclusive economic zone (darker shading). The inset illustrates the 50 km buffer construction using Ecuador's two EEZs as an example: one adjacent to the mainland coast and another surrounding the Galápagos Islands. For each EEZ, the area between the two outermost polylines represents the 50 km buffer used in our analysis.

4 Data

Study area. Our analysis focuses on Chinese fleet activity in 50 km buffers just outside foreign EEZs where Chinese vessels have no fishing access.[8] Chinese activity in these areas is close enough to matter strategically to the host country, which yet has no legal means to prohibit it. We exclude EEZs that grant Chinese vessels fishing access, since activity just outside them may reflect spillover from licensed fishing rather than strategic positioning.[9] We also exclude joint-regime areas and EEZs with overlapping claims, which lack a clearly defined host state. Because no public database tracks bilateral fisheries access agreements (Pauly et al., 2014), we infer access from observed vessel activity. We retain countries for which total Chinese fishing hours in the buffers over the sample period (i) are more than twice those inside their EEZs and (ii) exceed 1,000, as activity below these thresholds is more likely to reflect noise than deliberate positioning.[10] The final sample comprises 71 EEZs of

[8] Results are robust to buffer widths from 10 to 100 km (see Section 6.3 and Appendix Figure C.4).

[9] Many coastal states grant foreign fleets fishing rights in their EEZs in exchange for financial compensation (see e.g., Belhabib et al., 2015, 2020).

[10] Few countries lie near the ratio threshold: any threshold between 1.4 and 2.6 changes the sample by at most one country (Equatorial Guinea, ratio 1.93). Buffer fishing hours vary more continuously across countries; the 1,000-hour threshold falls between Norway (845 hours) and Mauritius (1,431 hours), the largest gap below 1,500 hours, just above a dense cluster of four countries with 758–845 hours.

21 countries, mapped in Figure 1 and listed in Appendix Table B.1. We vary the sample construction rules and show that our results are robust to these alternatives in Section 6.3 and Appendix Table C.7.

Fleet presence and fishing efforts. We use vessel activity data from Global Fishing Watch (GFW), which is reported at daily frequency since 2012 on a $0.1° \times 0.1°$ grid (about 11 km $\times$ 11 km at the equator) (Kroodsma et al., 2018; Global Fishing Watch, 2024). For each cell-day, GFW reports the presence and fishing hours of each detected vessel, identified by its unique Maritime Mobile Service Identifier (MMSI). We link activity with vessel characteristics, e.g., flag state and gear type, based on MMSI and aggregate the presence and fishing hours of all Chinese-flagged vessels across the grid cells in each 50 km buffer by month. At the extensive margin, our outcomes indicate whether any Chinese presence or active fishing is recorded in a buffer-month. At the intensive margin, we use total presence and fishing hours and the number of Chinese vessels present. Appendix Figure B.2 shows increasing trends of presence and fishing hours for all vessels and for Chinese vessels recorded in the buffers outside the analyzed EEZs, which reflect both genuine increases in DWF and likely expansion in GFW data collection. We are careful to note that GFW infers vessel activity from automatic identification system (AIS) signals, which vessels can deliberately disable mid-trip.[11] If vessels are more likely to hide their activity near politically distant hosts, intensive-margin outcomes may understate true deployment and bias our estimates downwards. Nevertheless, extensive-margin outcomes are less prone to this, as they capture only whether any hours are recorded rather than the precise count.

Geopolitical distance. Our main measure is the ideal point distance introduced by Bailey et al. (2017). This yearly measure is derived from voting patterns in the UN General Assembly (UNGA) and has become the standard measure of geopolitical distance between countries in both economics and political sciences (e.g., Kersting and Kilby, 2016; Clark and Dolan, 2021; Aiyar et al., 2024; Voeten, 2026). Its key methodological contribution is to hold the UN agenda constant, so that movement in a country's estimated position reflects changing alignment rather than shifts in topics at the UNGA. In essence, the ideal points locate each state relative to the US-led liberal international order. We define the political distance between each EEZ host country and China to be the absolute difference between their ideal points in a given year. This captures time-varying divergence between the host state and

[11] GFW applies convolutional neural networks to AIS signals recorded by satellite and terrestrial receivers to infer vessel characteristics and activities. AIS is mandated by the International Maritime Organization to avoid collision among vessels and is increasingly adopted by vessels operating on the high seas (Kroodsma et al., 2018). This reliance creates two caveats about GFW data. First, not all vessels carry AIS, although Kroodsma et al. (2018) argue that AIS captures most fishing activity on the high seas. Second, vessels can disable AIS to conceal their movements (e.g., Welch et al., 2022).

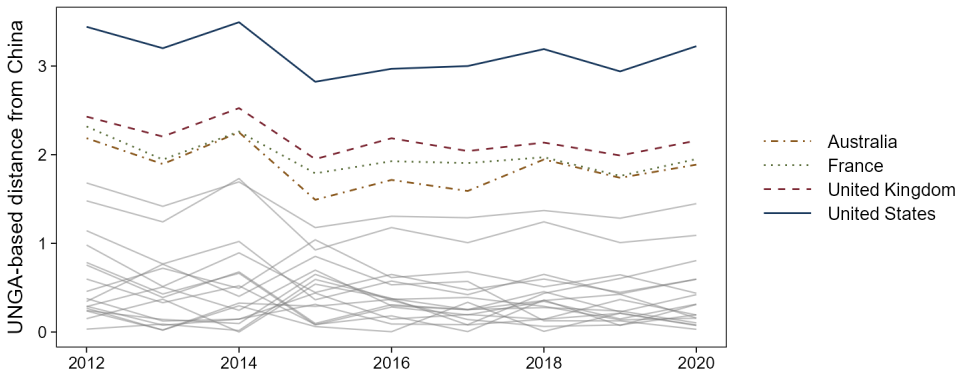


Figure 2: Ideal point political distance
Figure shows absolute difference in annual UN General Assembling voting-derived ideal points between host countries and China from 2012 to 2020.

China over how global issues should be resolved.[12] The resulting measure depicted in Figure 2 shows that, on average, the US exhibits the largest distance from China, followed by other western countries like France, the UK, and Australia. We also follow Davis et al. (2019) by constructing an event severity measure that utilizes incidents between governments from the GDELT data set (Leetaru and Schrodt, 2013). Each event is weighted by its severity, e.g., a border clash is weighted more heavily than a diplomatic denunciation. Importantly, we only consider negative events to capture diplomatic transgressions. The major difference between the index of Davis et al. (2019) and ours is that our measure is monthly instead of yearly. In the Appendix B.2, we detail how this measure is constructed and how it relates to the more established political distance measure.

Economic incentives. To control for the economic incentives driving distant-water fishing, we use two measures. First, we include the total fishing effort of all non-Chinese fleets in a given buffer-month. Second, we draw on phytoplankton abundance to proxy for marine life. We use MODIS satellite data (NASA), which measure ocean color reflectance and filter for the wavelength emitted by phytoplankton, yielding a measure of biological productivity at 9 km $\times$ 9 km resolution. This approach is similar to that of Flückiger and Ludwig (2015), with the key difference that our buffers are farther offshore and hence likely less productive than the continental-shelf areas studied in their setting.

Summary statistics of our data are reported in Appendix Table B.2. In Appendix B.3, we also provide detailed description of the complementary data sources used throughout the paper, including boundary shapefiles, additional ecological and economic controls, alternative

[12] As shown in Appendix C.3, our main results are robust when using simple UN (dis)agreement scores and, in some specifications, when using a severity score for diplomatic transgressions, indicating that our results are driven by a broader concept of geopolitical relations.

political-distance measures, and data employed in the mechanism analysis.

Descriptive evidence. As a preliminary step, we visualize the correlation between political distance from China and Chinese DWF activity just outside host EEZs. Figure 3 presents binned scatter plots after partialling out EEZ and time fixed effects. Panel (a) shows a positive correlation between political distance and the probability of Chinese fleet presence. This suggests that, conditional on host country and EEZ time-invariant characteristics and common time shocks, Chinese fleets are more likely to operate near an EEZ as its host country's political distance from China rises. Panel (b) reveals a similar pattern for active fishing. We provide the econometric estimates of these patterns in the next sections.

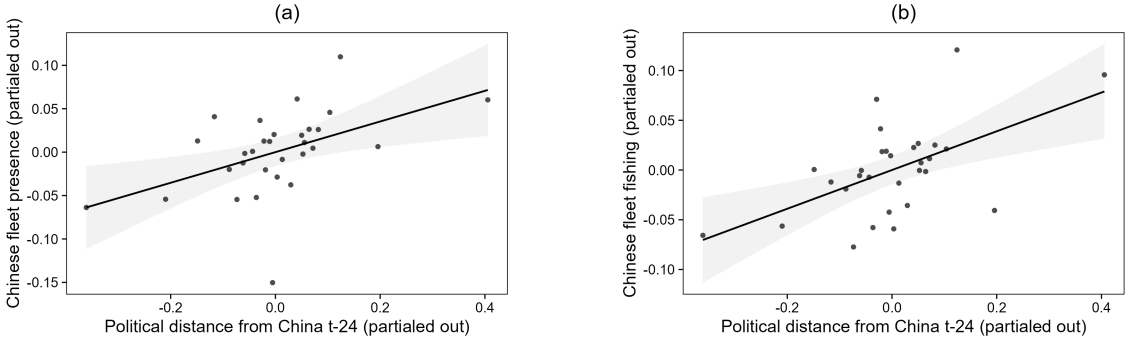


Figure 3: Correlation between political distance and Chinese fleet activity
Figure presents binned scatter plots of Chinese distant-water fishing (DWF) activity within 50 km outside host countries' EEZs against political distance from China, measured as the absolute difference in UN General Assembly ideal points. Panels (a) and (b) show the probability of fleet presence and active fishing, respectively. Both outcomes and political distance are residualized with respect to EEZ and year-month fixed effects. Fitted regression lines and corresponding 95% confidence intervals are shown.

5 Empirical framework

We estimate the following regression model to examine the role of political incentives in driving distant water fishing:

$$y_{i,c,t} = \beta PoliticalDistance_{c,t-24} + \varphi' \mathbf{X}_{i,t} + \alpha_i + \alpha_t + \alpha_{i,m(t)} + \varepsilon_{i,c,t}. \tag{7}$$

The main outcome, $y_{i,c,t}$, is an indicator for the presence of any Chinese vessel in the 50 km buffer outside the exclusive economic zone i of host country c in month-year t. As presence may also reflect vessels in transit, we use a separate indicator for active fishing, which captures whether any Chinese vessel engages in apparent fishing in the buffer. We estimate these extensive-margin specifications using linear probability models. $PoliticalDistance_{c,t-24}$ measures the absolute difference between the ideal points of the host country and China, derived from their UNGA votes (Bailey et al., 2017). Since ideal points are estimated by

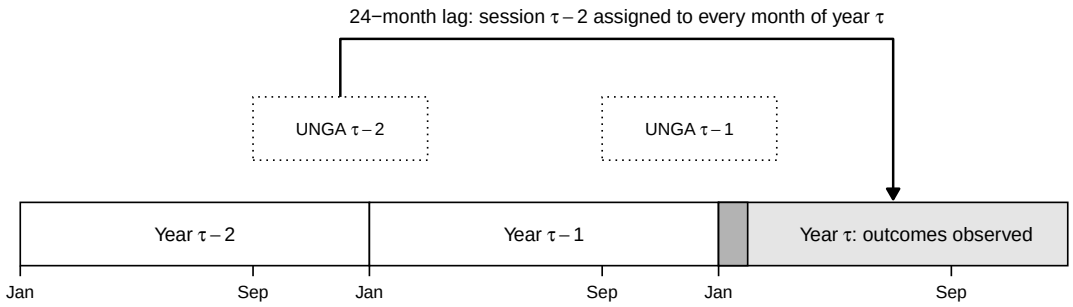


Figure 4: UN General Assembly voting timeline
Figure shows the UNGA sessions lasting from September to December of a given year and potentially into the early months of the following year. Lagging political distance by 24 months assigns the DWF outcome of every month of year τ to the ideal points derived from the UNGA session in year $\tau - 2$, which had fully ended before year τ began. Lagging by only 12 months would assign the early months of year τ (e.g., January as darker shaded) to a session whose votes may not yet be complete.

annual UNGA sessions, with the 24-month lag, an outcome observed in month t of a given year τ is assigned to the ideal-point distance from session $\tau - 2$. As illustrated in Figure 4, UNGA sessions typically last from September to December of a given year and may continue into the next year. A 12-month lag would therefore assign the outcome of the early months to a session which may not have concluded, while the 24-month lag ensures that political distance is fully measured before the outcome. The longer lag therefore helps rule out contemporaneous feedback from vessel activities to UNGA voting and also allows time for political dynamics to translate into fleet deployment decisions.[13]

To account for the economic incentives driving Chinese fleet activity, Eq. 7 controls for (i) the concurrent fishing effort of all non-Chinese fleets in the buffer and (ii) marine productivity, proxied by total phytoplankton abundance in the buffer 12 months prior.[14] These time-varying buffer characteristics are subsumed in the control vector $\mathbf{X}$. EEZ fixed effects α_i absorb time-invariant characteristics of the EEZ that affect the Chinese fleet presence in its buffer, including geographic location, climate, and the host country's time-invariant enforcement capacity and historical relations with China.[15] EEZ-month fixed effects $\alpha_{i,m(t)}$ control for the EEZ-specific seasonality of vessel presence and fishing. Time fixed effects α_t capture shocks common to all EEZs, such as increases in fleet subsidies, advances in marine technology, and upgrades in GFW data collection. They further account for common shifts in voting-based distance between China and other countries. Because political distance varies at the host-country level, we cluster standard errors by host country. Given the small num-

[13] We find consistent estimates when using a 18-month lag.

[14] The 12-month lag approximates the time from initial phytoplankton blooms to harvestable fish stock. The process from phytoplankton feeding to zooplankton to fry or juvenile fish varies across species but generally takes several months (see e.g., Schmidt, 2005; Michigan Sea Grant).

[15] Because EEZs are assigned to a single host country, EEZ fixed effects factor in time-invariant country-level controls.

ber of countries ($n = 21$) and the uneven number of EEZs per country, for every estimation we also report the wild cluster bootstrap p-value based on Cameron et al. (2008).[16]

β captures the change in Chinese DWF fleet activity associated with a one-point increase in UNGA voting-based political distance between the host state and China. This corresponds to roughly one standard deviation of political distance in our sample (see Appendix Table B.2), and is equivalent to moving from a typical European (France, UK) voting position to that of the United States (see Figure 2). Our identifying assumption is that, conditional on location fixed effects, common trends, and economic incentives, variation in political distance is uncorrelated with unobserved determinants of DWF fleet activity absorbed in the error term. Two primary threats could violate this assumption. The first is reverse causality: persistent Chinese fleet activity just outside an EEZ border could itself contribute to deteriorating bilateral relations and shape future UN voting. Although our use of lagged political distance mitigates this concern, we estimate a distributed-lag model to directly assess it. The second is omitted variable bias: unobserved factors that affect both political distance and fleet activity could confound our estimates. We address this through a series of tests that progressively layer in controls for ecological conditions and bilateral economic ties. To jointly mitigate endogeneity concerns, we further implement a 2SLS approach exploiting the issue structure of the UNGA votes. Lastly, we examine the sensitivity of our estimates to alternative choices in data construction. The results are discussed in Sections 6.2 and 6.3.

6 Political distance and fleet activity

In this section, we first present results on the extensive-margin relationship between political distance and the likelihood of fleet presence and fishing activity. We then address concerns regarding the validity of the estimates through a range of complementary specifications and robustness checks. Finally, we turn to the intensive margin and examine how political distance is associated with the scale of fishing effort and vessel deployment.

6.1 Extensive margin

We begin with the extensive-margin results in Table 1. In columns (1)-(3), the dependent variable is an indicator for whether Chinese DWF presence is recorded within the 50 km buffer just outside a given EEZ in a given month. Column (1) reports a parsimonious specification that controls only for the sea-route distance between China's EEZ and the host

[16] Throughout, we implement the wild cluster bootstrap using the `fwildclusterboot` R package with Rademacher weights and 9,999 replications.

Table 1: Political distance and extensive margin of fleet presence and fishing

	1(CHN fleet presence)			1(CHN fleet fishing)		
	(1)	(2)	(3)	(4)	(5)	(6)
Political distance $_{t-24}$	-0.090*	0.177**	0.149**	-0.103***	0.191**	0.167*
	(0.050)	(0.076)	(0.072)	(0.035)	(0.092)	(0.091)
	[0.071]	[0.014]	[0.045]	[0.005]	[0.035]	[0.084]
Log sea-route distance	-0.327			-0.157		
	(0.099)			(0.072)		
	[0.105]			[0.124]		
Log non-CHN fishing hours			0.026***			0.024***
			(0.003)			(0.004)
			[0.000]			[0.001]
Log phytoplankton $_{t-12}$			0.035			0.022
			(0.020)			(0.023)
			[0.117]			[0.372]
EEZ FE	–	✓	✓	–	✓	✓
Month-year FE	–	✓	✓	–	✓	✓
EEZ-month FE	–	–	✓	–	–	✓
Mean dependent variable	0.368	0.368	0.368	0.238	0.238	0.238
Observations	7,668	7,668	7,668	7,668	7,668	7,668

Notes: 'CHN fleet presence' is an indicator variable equal to one if any presence hour is recorded for the Chinese fleet in the buffer, and zero otherwise. 'CHN fishing presence' is an indicator variable equal to one if any fishing hour is recorded for the Chinese fleet in the buffer, and zero otherwise. 'Political distance' is the absolute difference in UNGA voting-based ideal point between China and the EEZ's host country two years prior. 'Log sea-route distance' is the logarithmized sea-route distance from the centroid of China's EEZ to the centroid of the host EEZ. 'Log non-CHN fishing hours' is the logarithmized number of fishing hours recorded for all non-Chinese fleets in the buffer. 'Log phytoplankton' is the logarithmized amount of phytoplankton abundance measured for the buffer one year prior. Standard errors clustered at the country level are reported in the parentheses. Wild cluster bootstrap p-values are in square brackets, clustered at the country level. *p<0.1; **p<0.05; ***p<0.01, based on wild cluster bootstrap p-values.

EEZ, which accounts for the average travel costs. Without location and time fixed effects, the between-EEZ estimates show that a one-point increase in political distance between the host country and China is associated with a 9 percentage point decrease in the probability of observing Chinese fleet presence. In addition, a 10 percent increase in sea-route distance is associated with a 3 percentage-point decrease.[17] The negative coefficient on political distance partly reflects that host countries with greater political distance from China, as illustrated in Figure 2, tend to be both geographically farther and higher-income, thus likely possess stronger enforcement capacity. These factors in turn may reduce the probability of the Chinese DWF presence at the border.

In column (2), when we account for EEZ and time fixed effects, the coefficient on political distance becomes positive. The estimate implies that, controlling for host country and

[17]With standard errors clustered at the country level as reported, the coefficient on distance is close to being statistically significant at the 10% level. Since distance is measured between China and each EEZ, when we cluster standard errors at the EEZ level, the coefficient becomes statistically significant at the 1% level. Results are reported in Appendix Table C.8.

EEZ time-invariant characteristics and global shocks, including common changes in voting position, a one-point increase in political distance between the host country and China is predicted to raise the probability of Chinese fleet being present in the buffer by 17.7 percentage points. In column (3), we further include EEZ-specific seasonality fixed effects and time-varying controls for the economic incentives for fishing activity. The coefficients on the controls imply that Chinese DWF presence is stronger in places where non-Chinese fleets are also fishing and where marine productivity is higher.[18] Importantly, the coefficient on political distance is only modestly attenuated and remains positive and statistically significant, indicating that the within-EEZ association is robust to controlling for time-varying economic drivers of fishing effort. In this specification, a one-point increase in political distance is estimated to raise the probability of the Chinese DWF presence by 14.9 percentage points, approximating 40% the outcome mean.

Columns (4)-(6) repeat the specifications in the previous columns using an indicator for whether the Chinese DWF fleet is actively fishing in the buffer. This outcome speaks more directly to the political-economy mechanism in our framework, since presence measure could also capture vessels in transit. The pattern of estimates mirrors columns (1)-(3). The cross-sectional association between political distance and the probability of fishing is negative, but turns positive once time-invariant EEZ characteristics and global trends are absorbed. The within-EEZ effect is also larger for this outcome. In the full specification of column (6), a one-point increase in political distance is predicted to raise the probability of Chinese DWF fishing in the buffer by 16.7 percentage points, about two thirds of the outcome mean.

6.2 Threats to identification

Our identification rests on the assumption that, conditional on time-invariant EEZ characteristics, global common shocks, and time-varying controls proxying for the economic incentives for fishing, UNGA voting-based political distance between the host country and China is uncorrelated with unobserved determinants of the Chinese DWF activity just outside the EEZ. Two threats could violate this assumption. First, the estimates could be biased by reverse causality. Specifically, pronounced activity of the Chinese fleet just outside an EEZ border may itself worsen bilateral relations and shift future voting. Second, omitted variables which jointly influence China-host political distance and the Chinese DWF activity could bias the baseline estimates. A further concern is that our results are only valid when UNGA voting indeed captures geopolitical divergence. In this section, we present various complementary

[18] Like sea-route distance, the phytoplankton abundance is measured at the EEZ buffer-month level. In Appendix Table C.8, the coefficient on phytoplankton becomes more precisely estimated when standard errors are clustered at the EEZ instead of the country level.

approaches to address these concerns.

Reverse causality. A natural question is whether stark presence of a foreign DWF just outside an EEZ also deteriorates bilateral relations. If so, the baseline specification and the estimates in Table 1 would partly reflect reverse causality rather than the effect of China-host political distance on the Chinese DWF activity. Although lagging the voting-based distance by 24 months mitigates reverse causation, it remains an issue if this lagged measure is correlated with contemporaneous and future voting. To better assuage this concern, we estimate a distributed-lag specification that jointly includes one-year and two-year leads and lags of political distance, alongside its contemporaneous value (Appendix Eq. C.1). The results reported in Appendix Figure C.1 in fact do not provide support for a reverse-causality interpretation. Only the two-year lag of China-host political distance loads positively and significantly on the Chinese DWF presence and fishing probability, in line with the baseline results. In contrast, the coefficients on neither the leads nor the contemporaneous measure of political distance yield positive estimates as would be predicted by reverse causality.

Potential confounders. The second threat to identification is that factors jointly affecting political distance and DWF activity may be omitted from our estimation. We address this concern in two steps. First, we successively add ecological and economic controls to the baseline specification: (i) anomalies of sea-surface temperature, which regulates the growth of marine species populations (Wang et al., 2020; van der Sleen et al., 2022), and the occurrence of cyclones, which can interfere with DWF activity; and (ii) economic ties between the host country and China, proxied by bilateral trade volume and the existence of a trade agreement, together with host country's income as a measure of enforcement capacity. Results in Appendix Table C.1 show that the coefficient on political distance is qualitatively and quantitatively stable as additional controls are layered in, suggesting that the baseline estimates are unlikely to be driven by these factors. Second, we conduct a placebo test using the two foreign DWF fleets most comparable to China's in scope and economic incentives—the Taiwanese and Japanese fleets (see Appendix Figure C.2). If our baseline reflects general economic or regulatory determinants of DWF, China-host political distance should also predict the presence and fishing of these fleets. This is not the case. The coefficients reported in Appendix Table C.2 are negligible in magnitude and statistically insignificant, indicating that the baseline estimates reported in Table 1 captures a China-specific relationship rather than economic or regulatory factors that shape DWF in general.

Measuring political distance. A further concern is whether the UNGA voting-based measure truly captures geopolitical distance. We address two specific issues pertaining to this. First, the results may reflect components of the ideal point that do not capture actual

variation in interstate relations. To test this, we re-estimate the baseline using alternative measures of political distance from the literature: vote agreement scores, two measures constructed from the UNGA DM dataset by Fjelstul et al. (2023, 2025), together with the GDELT event-based severity measure. Appendix Table C.3 shows that the coefficient on political distance remains positive and statistically significant under nearly all alternatives. Second, UNGA votes may not reflect genuine foreign-policy preferences in the presence of vote buying. Donors such as China may reward favorable votes with aid and trade concessions (e.g., Kuziemko and Werker, 2006; Dreher et al., 2008, 2009, 2018). In this case, dependent hosts would vote more similar to China and also tolerate Chinese DWF activity at their border more. As this mechanism implies *increased* Chinese fleet activity when political distance becomes *smaller*, we first note that vote buying would predict the opposite of our finding. As a direct check, Appendix Table C.4 shows that our results are driven by host countries least reliant on China in both trade and aid, hence least exposed to vote buying. Taken together, these exercises support the interpretation that our baseline estimates reflect geopolitical divergence rather than artifacts of the baseline measure or strategic voting.

Two-stage least squares. To address the remaining confounder concern and measurement error, we implement a two-stage least squares strategy that exploits the issue structure of the UNGA votes. Specifically, we instrument the baseline political distance with China-host vote disagreement on two issues classified by Voeten (2013): (i) nuclear weapons and nuclear materials proliferation and (ii) arms control and disarmament. The issues both target security-related topics and characterize divergence in geopolitical stances. Importantly, they have no plausible direct links with fishing access, trade, or maritime regulation. As fishing disputes are unlikely to affect votes on these issues, this approach also helps mitigate reverse causality. For each issue, we construct the IV to be the share of votes in a given session that the host country and China vote differently.[19] The results in Appendix Tables C.5 and C.6 show that vote disagreement on nuclear weapons and on arms control issues significantly predict the baseline political distance measure. Second-stage estimates using both instrumental variables are directionally similar to the OLS estimates, which imply increased probability of the Chinese DWF fleet presence or fishing when political distance widens. The estimate using nuclear weapons-issue disagreement is statistically significant at the 5% level, while the one using arms control is less precisely estimated. Notably, the 2SLS estimates using vote disagreement on nuclear weapons issue as IV are about 0.3, larger than the OLS esti-

[19]The instruments use exact vote mismatches on roll calls in specific issue domains, following the agreement-score approach of Signorino and Ritter (1999). Details are presented in Appendix C.4. By contrast, the baseline political-distance measure of Bailey et al. (2017) estimates a one-dimensional latent ideal point using a dynamic item-response model fitted to votes across the full set of UNGA roll calls.

mate. One possible explanation for this difference is that the 2SLS estimates capture a local average treatment effect, identified among host countries whose baseline political distance from China is most responsive to variation in voting disagreement on the nuclear weapons issue. Taken together, these patterns provide evidence supporting that our baseline captures political distance driven by geopolitical divergence rather than confounding economic or fisheries-specific regulatory misalignment.

6.3 Robustness

We complement the identification tests with robustness checks assessing the sensitivity of our baseline estimates to sample composition, outcome construction, and inference approach.

Sample composition. Because our sample is selected based on Chinese fishing activity within the EEZs and their buffers, which is closely related to our outcome, we vary the sample restriction thresholds described in Section 4. The results in Appendix Table C.7 show that relaxing these restrictions sometimes reduces estimation precision, likely because it adds countries that allow Chinese vessels to fish inside their EEZs or rarely experience Chinese vessel activity near their boundaries. Nonetheless, across alternative constructions, the estimated coefficients on political distance remain positive in sign and quantitatively comparable to the baseline results. We also re-estimate the baseline excluding one host country or EEZ at a time. The coefficients on political distance reported in Appendix Figure C.3 range from 0.1 to 0.15 (for probability of presence outcome) and from 0.1 to roughly 0.2 (for probability of fishing outcome). These results indicate that no single host country or EEZ drives the results in Table 1.

Buffer width. Our outcomes at the baseline measure activity of the Chinese DWF fleet within 50 km outside an EEZ boundary. In Appendix Figure C.4, varying the buffer distance from 10 to 100 km yields estimates that are similar to the baseline. We observe that the coefficient magnitudes rise with the buffer width, as the probability of Chinese fleet presence or fishing likely increases with the area covered.

Inference. Our main specification clusters the standard error at the host country level and applies the wild cluster bootstrap to account for the small number of clusters. In Appendix Figure C.5, we provide the inference results for the coefficients on political distance based on various clustering level (EEZ, country×year, two-way by country and year) and spatially robust standard errors based on Conley (1999). Throughout, the coefficients on political distance remain statistically significant at the 5% and 10% levels. As a final check, we conduct a permutation test in which political distance is randomly reassigned across host

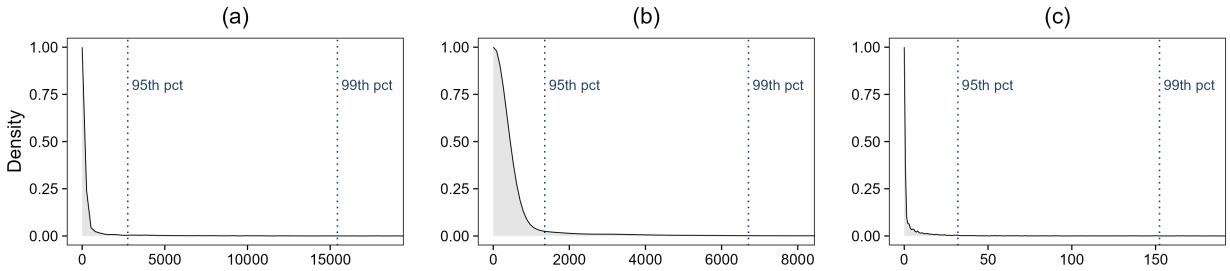


Figure 5: Distributions of intensive-margin outcomes
Figure plots the distributions of the main intensive-margin outcomes of the Chinese DWF fleet's total presence hours (Panel a), total fishing hours (Panel b), and number of unique vessels present (Panel c) across the EEZ buffer-months in the sample. In each panel, vertical dotted lines mark the 95th and 99th percentiles.

countries within each month-year and re-estimate the baseline specification on each permuted sample. Appendix Figure C.6 shows that the coefficients across 5,000 permutations are centered on zero and far smaller than the actual estimates reported in Table 1, indicating that the baseline relationship is unlikely to arise by chance.

6.4 Intensive margin

In the previous sections, we document that rising political distance between China and a host country is predicted to increase the probability of the Chinese DWF fleet being present or fishing just outside the host's EEZ. We turn to explore the intensive margin by looking into three outcomes: the total number of presence hours recorded for the Chinese fleet, an similar measure for active fishing, and the total number of Chinese vessels present. As illustrated in Figure 5, these count outcomes contain many zeros and are highly right-skewed. We thus employ the Poisson pseudo-maximum-likelihood (PPML) estimator in the baseline (see e.g, Santos Silva and Tenreyro, 2006).[20,21] As a robustness check, we additionally estimate the model with OLS using the log of the outcome, restricting the sample to buffer-months with positive observations.

Table 2 presents the results. Panel A reports results for the outcome of total presence hours.

[20]The PPML model takes the form:

$$y_{i,c,t} = \exp(\beta^P PoliticalDistance_{c,t-24} + \vartheta' \mathbf{X}_{i,t} + \alpha_i^P + \alpha_t^P + \alpha_{i,m(t)}^P)\, v_{i,c,t},$$

where outcome y is the intensive-margin measures. The included variables are similarly defined as in the main linear probability model in Eq. 7. Throughout, we cluster the standard errors at the host country level and base inference on the leave-one-cluster-out jackknife standard errors to correct for the small number of cluster in the PPML models, for which the wild cluster bootstrap is not directly available (see e.g., MacKinnon et al., 2023, 2025).

[21]Because singleton observations within fixed-effect groups whose outcomes contain only zeros are dropped from the estimation, the sample size is smaller than in the extensive-margin analysis.

Table 2: Political distance and intensive margin of fleet presence and fishing

	(1)	(2)	(3)	(4)	(5)
Estimator	PPML	PPML	PPML	PPML	OLS
Outcome	Raw	Wins. 99%	Wins. 95%	Raw	log(y)
Sample	Full	Full	Full	Excluding struct. hosts	Positives only
	Panel A. Outcome is total number of CHN fleet presence hours				
Political distance $_{t-24}$	-0.232	0.808**	0.952***	0.616**	0.648
	(0.340)	(0.220)	(0.187)	(0.204)	(0.383)
	[0.777]	[0.012]	[0.000]	[0.031]	[0.145]
Mean dependent variable	1,403.7	804.7	389.4	283.3	5.3
Observations	4,770	4,770	4,770	3,879	2,738
	Panel B. Outcome is total number of CHN fleet fishing hours				
Political distance $_{t-24}$	0.132	0.842**	0.886***	0.498	0.818
	(0.352)	(0.284)	(0.248)	(0.264)	(0.495)
	[0.838]	[0.025]	[0.005]	[0.180]	[0.156]
Mean dependent variable	710.6	456.2	221.2	235.7	5.1
Observations	3,879	3,879	3,879	2,993	1,737
	Panel C. Outcome is total number of CHN vessels present				
Political distance $_{t-24}$	0.334***	0.499***	0.768***	0.363***	0.257
	(0.097)	(0.114)	(0.142)	(0.096)	(0.129)
	[0.008]	[0.003]	[0.000]	[0.007]	[0.102]
Mean dependent variable	12.0	11.0	6.46	4.83	2.0
Observations	4,770	4,770	4,770	3,879	2,738
Controls	✓	✓	✓	✓	✓
EEZ FE	✓	✓	✓	✓	✓
Month-year FE	✓	✓	✓	✓	✓
EEZ-month FE	✓	✓	✓	✓	✓

Notes: Outcomes in Panels A-C are the total number of presence hours, fishing hours, and unique vessels recorded for the Chinese fleet in a given EEZ buffer-month, respectively. 'Political distance' is the absolute difference in UNGA voting-based ideal point between China and the EEZ's host country two years prior. Col. (1) reports Poisson pseudo-maximum-likelihood (PPML) estimates on the full sample. Cols. (2)-(3) winsorize the outcome at the 99th and 95th percentiles, respectively. Col. (4) excludes structural-host EEZs (Argentina, Chile, Ecuador, Peru, Russia, and Japan). Col. (5) uses OLS with the outcome in logs, restricting the sample to EEZ buffer-months with positive outcomes. The PPML estimator (Cols. 1-4) drops singleton observations within fixed-effect groups whose outcomes are always zero, reducing the sample below the full sample used in the extensive-margin analysis. Controls are log number of fishing hours recorded for non-Chinese fleets in the buffer and log amount of phytoplankton abundance measured for the buffer one year prior. Standard errors clustered at the country level are reported in the parentheses. Square brackets report p-values based on leave-one-cluster-out jackknife standard errors at the country level. *p<0.1; **p<0.05; ***p<0.01, based on CV3 p-values.

The coefficient in column (1) is negative, suggesting that increasing political distance between China and host is associated with a decrease in the total number of Chinese fleet presence hours in the 50 km buffer. The estimate is yet not statistically significant. The imprecision likely reflects the high variance and right skew of the outcome, whereby a small number of observations with extremely large presence hours could disproportionately influence the estimate. We probe this along several margins. We first winsorize the outcome

at its 99th and 95th-percentile values in columns (2) and (3). Column (2) shows a positive and statistically significant coefficien. A one-point increase in the China-host political distance is predicted to increase the number of the Chinese fleet being present in the buffer by about 124%.[22] The pattern is similar in column (3).[23] The contrast between column (1) and columns (2)-(3) suggests that the extremely large outcome observations may be concentrated in hosts where Chinese fleet activity at the border is motivated by structural factors other than political distance. The six hosts with outlier presence hours are Ecuador, Peru, Chile, Argentina, Russia, and Japan. The Southern American coastal nations host highly productive marine ecosystems and are favored fishing grounds for many DWF fleets including China's, whereas Russia and Japan are the neighbor countries with whom China shares long-term fisheries agreements.[24] For these reasons, the Chinese DWF activity next to waters of these hosts is probably less responsive to political distance, which may introduce noise in the full sample estimation. To test this, column (4) drops the six structural hosts and retains the outcome in its raw form. The coefficient on political distance is positive and statistically significant, implying that for a *non*-structural host, a one-point increase in political distance from China is associated with roughly 85% higher Chinese fleet presence hours in the buffer. In column (5), we employ OLS on the natural log of presence hours, essentially restricting the sample to buffer-months with positive presence. The point estimate is comparable to those in columns (2)-(4), albeit less precisely estimated, suggesting that among buffer-months with any Chinese fleet presence, greater political distance is associated with more intensive activity. Combined, the results suggest that the imprecise estimate in column (1) reflects influence from the extreme observations in structural-host EEZs. For the bulk of the remaining observations, increasing China-host political divergence is robustly associated with more Chinese fleet presence hours just outside the host's EEZ.

In Panel B, we repeat the exercise with the fleet's fishing hours. The pattern is qualitatively and quantitatively similar to Panel A. The coefficient on political distance is positive but not statistically significant on the raw full sample in column (1). However, it remains positive and increases in magnitude and precision once large observations in structural hosts are winsorized (columns 2-3) or excluded (column 4). Panel C (Table 2) looks into a third dimension of activity intensity: the number of unique vessels in the Chinese fleet present in a given buffer-month. Two features distinguish this outcome from the hours-based measures. First, the vessel count is bounded by the size of the Chinese DWF fleet (Figure 5, Panel

[22] $100 \times (\exp(0.808) - 1) = 124.3$.

[23] Truncating the sample at these tail values—rather than winsorizing—yields very similar point estimates.

[24] The Sino-Japanese Fisheries Agreement, originally signed in 1975, governs cooperative management in the East China Sea (see e.g., Keyuan, 2003). China and the Soviet Union signed a bilateral Agreement on Cooperation in Fisheries in 1988 (see Ministry of Foreign Affairs of the People's Republic of China, 1988).

c), so the outlier-influence problem driving the imprecise estimates in Panels A-B could be attenuated. Second, vessel counts are more robust to measurement error. While hours are reduced whenever a vessel temporarily switches off its AIS transponder mid-trip, the same vessel still typically appears in the buffer at some other point and remains in the count. In line with these advantages, column (1) yields a positive and statistically significant coefficient even on the raw outcome and full sample. It reveals that a one-point increase in China-host political distance is predicted to raise the number of Chinese vessels present in the host EEZ's buffer by about 39%. Winsorizing the outcome at extreme values or dropping structural hosts likewise produces statistically significant positive coefficients on political distance.

Together, the results presented in Table 2 provide evidence suggesting that Chinese DWF fleet's deployment just outside a host's EEZ intensifies as China-host political distance increases. This finding complements the extensive-margin results in Section 6.1 by showing that not only is the fleet more likely to be present outside the EEZs of more politically distant hosts, but it also registers more activity hours and has more vessels there. This signals that political incentives shape deployment beyond economic considerations. Specifically within our theoretical framework in Section 3, the empirical pattern aligns with political payoffs outweighing deterrence from the host state (net political gains), which potentially sustains presence and fishing effort even when economic returns are negative. We shed light on the specific political mechanisms underlying this pattern in Section 7.

7 Mechanisms

In Section 6, we document that the Chinese DWF activity just outside the EEZ increases as China-host political distance becomes larger. This implies that the Chinese DWF fleet presence and active fishing may yield strategic value that outweighs any drawback of increased enforcement and scrutiny by host countries' coast guards and navies. This section investigates the nature of these strategic payoffs. The empirical investigations draw on the theoretic considerations of Section 3.1, which presupposes power projection, surveillance, and retaliation as strategic motivations. Because the aggregate effects are consistent with multiple mechanisms, we proceed by systematically evaluating each channel separately. Alongside these China-driven channels, we also revisit the enforcement channel from the host country side. First, we examine heterogeneity at the fleet level, which points towards the surveillance channel as one with strongest empirical support. Afterwards, we leverage a rich set of vessel-level information to rigorously assess the plausibility of surveillance as a primary driving mechanism and to investigate the type of information these vessels may be gathering.

Table 3: EEZ Heterogeneity Analysis

	1(CHN fleet presence)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Political distance$_{t-24}$	0.198**	0.179*	0.130	0.230***	0.098	-0.102	
	(0.078)	(0.079)	(0.083)	(0.081)	(0.070)	(0.135)	
	[0.011]	[0.053]	[0.146]	[0.006]	[0.192]	[0.539]	
Political distance$_{t-24}$ × Far capital	-0.352						
	(0.045)						
	[0.135]						
Political distance$_{t-24}$ × Large EEZ		-0.107					
		(0.105)					
		[0.395]					
Political distance$_{t-24}$ × Strong navy			0.053				
			(0.109)				
			[0.652]				
Political distance$_{t-24}$ × Near China				-0.340*			
				(0.097)			
				[0.052]			
Political distance$_{t-24}$ × Small fleets					0.169**		
					(0.063)		
					[0.047]		
Political distance$_{t-24}$ × Democratic						0.402**	
						(0.142)	
						[0.049]	
Negative events$_{t-5,\ldots,t-1}$ (std.)							0.013
							(0.008)
							[0.567]
Controls	✓	✓	✓	✓	✓	✓	✓
EEZ FE	✓	✓	✓	✓	✓	✓	✓
Month-year FE	✓	✓	✓	✓	✓	✓	✓
EEZ × month FE	✓	✓	✓	✓	✓	✓	✓
Observations	7,668	7,668	7,668	7,668	7,668	7,668	7,668

Notes: 'CHN fleet presence' is an indicator variable equal to one if any presence hour is recorded for the Chinese fleet in the buffer, and zero otherwise. 'Political distance' is the absolute difference in UNGA voting-based ideal point between China and the EEZ's host country two years prior. 'Far capital' equals one if the marine distance from the EEZ to the host country's capital exceeds the 75th percentile. 'Large EEZ' equals one if the EEZ's area exceeds the 75th percentile. 'Strong navy' equals one if the host country's naval tonnage based on Crisher and Souva (2014) exceeds the 75th percentile. 'Near China' equals one if the marine distance from the EEZ to China is below the 25th percentile. 'Small fleets' equals one if the average size of the Chinese subfleet present in the EEZ's buffer is below the median. 'Democratic' equals one if the host country's mean score on the Economist Intelligence Unit (2024) over the estimation period 2012–2020 is at least 6. 'Negative events' is the sum, over the preceding four months, of the absolute Goldstein score of the most severe negative event per day directed by the host country's government or military at the Chinese government (See Appenndix B.2 for details). Controls are log number of fishing hours recorded for non-Chinese fleets in the buffer and log amount of phytoplankton abundance measured for the buffer one year prior. Standard errors clustered at the country level in parentheses. Standard errors clustered at the country level are reported in the parentheses. Wild cluster bootstrap p-values are in square brackets, clustered at the country level. *p<0.1; **p<0.05; ***p<0.01, based on wild cluster bootstrap p-values.

7.1 Fleet-level heterogeneity

To elucidate mechanisms underlying the increase in Chinese DWF activity, we first conduct a heterogeneity analysis at the fleet level using various EEZ and subfleet-specific characteristics for the outcome indicating the probability of the Chinese DWF fleet presence. The results are presented in Table 3. We provide the estimates for outcome indicating active fishing in Appendix Table D.1, which are consistent with those in Table 3 discussed below.

Enforcement. We first consider potential enforcement imposed by the host country. In detail, deteriorating relations may induce coastal states to increase monitoring intensity and enforcement actions against foreign vessels. Our baseline results already suggest that strategic payoffs likely exceed any increase in enforcement intensity when relationships worsen. Columns (1)-(3) of Table 3 further disentangles this average effect based on characteristics that determine enforcement capacities of host countries. In the first column, if enforcement plays an important role, EEZs far from the host country's seat of power should possess less policing capacity and therefore may allow Chinese vessels greater freedom to operate undetected. The interaction term, however, points to the opposite being the case. The estimates are imprecise, but if anything imply that the main effect comes from EEZs that are nearer to their capitals. This weakens the case for enforcement and, as elaborated upon later, strengthens the case for surveillance. Supporting this notion, we do not find stronger effects for larger EEZs (column 2), which may be more difficult for the host to monitor, or weaker effects for EEZs belonging to hosts with stronger navies (column 3). While we do not find evidence in favor of enforcement here, it does not mean that this channel is an irrelevant predictor of Chinese vessel movement. Within the time span of our sample, it is unlikely that enforcement capacity changes significantly for any country, thus making within-country variation less useful to capture this mechanism. The between-country results in Table 1 (column 1) suggest that EEZs of the most politically distant countries, i.e., the US and its allies, have fewer Chinese vessels at their borders, even after controlling for distance. These countries are more developed and hence likely have higher baseline enforcement capacity.

Power projection. We next examine whether presence of the Chinese DWF fleet is motivated by power projection. If this were the primary motive, we would expect stronger effects in EEZs closer to China, where security stakes are higher and strategic interests are most salient, notably the South China Sea and Taiwan. Evidence from Table 3 instead does not suggest this is the case. Column (4) shows that the baseline political-distance effect, in fact, attenuates in closer EEZs. We are careful to note that the sample excludes overlapping claims and thus coastal countries in the South China Sea. Therefore, while power projection is unlikely to act as the major motivating mechanism within our scope, this may not hold externally. Importantly, column (5) indicates that the effect is driven by EEZs where the average size of the visiting Chinese DWF fleet is smaller. This pattern is inconsistent with power projection, which would typically involve a large, visible fleet presence rather than small-scale deployments. Figure 6 reinforces this notion. Panel (a) illustrates that our political distance effect makes it more likely for small fleets to appear. The coefficient is largest and statistically significant for very low vessel-count thresholds and turns negative once fleet size exceeds 40. Finally, Panel C of Table 2 indicates that the magnitude of this

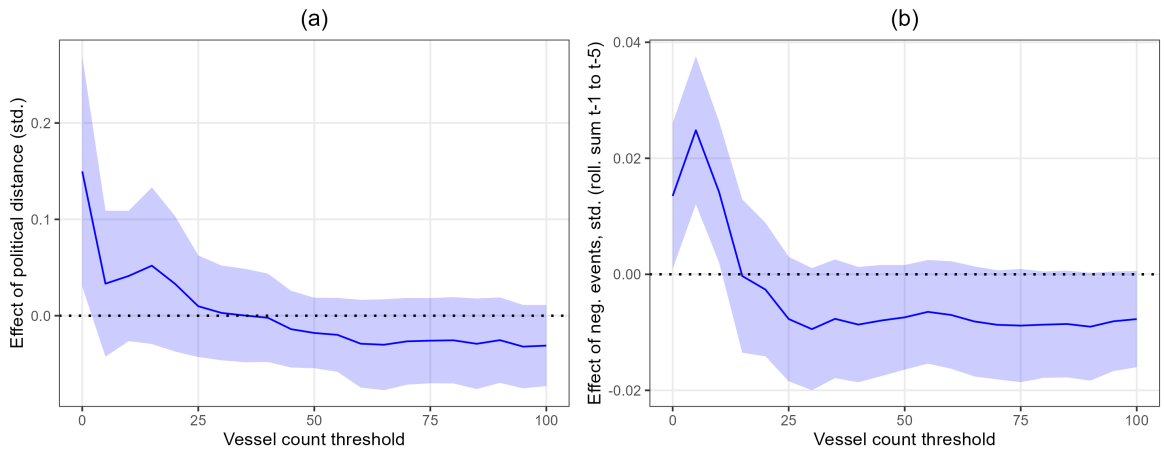


Figure 6: Effect of political distance and negative events by fleet size
Figure presents the probability of Chinese DWF presence with fleets above ship count thresholds between 0 and 100. Every fifth threshold is estimated via an independent regression. Panel (a) shows the effect of our main political distance variable on the chance of a fleet above each threshhold appearing, panel (b) shows the effect of the severity-weighted rolling sum of negative events in $t-1$ until $t-5$ based on our GDELT measure. Only events in which China is the object of an action are considered in (b). Both panels show effect estimates and the corresponding 95% confidence interval and with controlls similar to the main regressions.

effect is modest: a one-standard-deviation increase in political distance is predicted to bring only about five additional vessels to the buffer on average.

Surveillance. The surveillance channel conjectures that deteriorating interstate relations increase incentives to monitor the foreign countries, particularly through inconspicuous, smaller fleets that provide just enough cover and support without drawing substantial attention. The heterogeneity patterns in Table 3 are most consistent with this interpretation. As aforementioned, column (5) shows that the baseline estimate is driven by fleets of small sizes, aligning with low-visibility monitoring rather than large-scale signaling. Moreover, column (6) demonstrates that effects are concentrated in EEZs belonging to democratic host countries, which are more likely to be embedded in Western security networks and therefore present higher intelligence value. A further argument for surveillance rather than power projection is that fishing activity itself, not only presence, rises with political distance (Tables 1 and 2). [25] Overt power projection would not require fishing as cover. Surveillance, by contrast, plausibly does, since fishing provides a commercial rationale for sustained vessel presence just outside a foreign EEZ.[26]

[25]Heterogeneities for smaller fleets and democratic host countries are also statistically significant with fishing as the outcome variable in Appendix Table D.1, further indicating that our effects are not only driven by presence.

[26]A prominent example of overt projection is the coordinated massing of Chinese fishing vessels documented by Buckley et al. (2026), in which a giant wall of vessels intended to demonstrate the capability and coordination of the fleet to block ships from neighboring states. During this operation, Chinese fishing vessels were not observed to fish.

Retaliation. Finally, we consider retaliation. This mechanism requires that responses be timely, observable, and attributable to specific diplomatic transgressions. It therefore predicts short-run increases in vessel activity following salient disputes, potentially accompanied by economically costly fishing pressure. To probe this possibility, we utilize our GDELT event-based measure of diplomatic transgressions in order to test the effect of the cumulative severity of bilateral disputes within the preceding 5 months, excluding the current one. The last column in Table 3 reveals a positive estimate, which however is not statistically significant.[27] Figure 6, Panel (b), further shows that negative diplomatic events are associated with an increase in small fleets and a decline in large fleets. This is a similar pattern to our baseline political distance measure in Panel (a) and inconsistent with coercive retaliation as smaller fleets will only have limited punitive economic impact. While this is suggestive of some responsiveness to diplomatic shocks, the overall pattern is again more consistent with heightened monitoring following diplomatic tension rather than punitive escalation. The negative effects on larger fleet sizes can be interpreted as reflecting either increased host country enforcement or deliberate restraint by China to avoid escalation. In line with this, we find an even stronger negative reaction, also among smaller fleets, when we examine the thresholds for number of vessels fishing in Appendix Figure D.1. Fishing is likely even more relevant for diplomatic repercussions due to inuring economic costs onto host countries. We thus reject retaliation as the main strategic motive and document the opposite effect, with larger fleets becoming less common shortly after interstate conflicts.

7.2 Vessel-level heterogeneity

In the previous section, we find that heterogeneity patterns at the fleet level are consistent with surveillance as the primary underlying motive. Nevertheless, the analysis still masks which vessels conduct the activity or what kind of information they may collect. We therefore turn to vessel-level characteristics, both to further test the surveillance interpretation and to characterize the behavior consistent with it.

Our method in this section is guided by the idea that if the vessels behind our results position themselves at the border for surveillance, their characteristics and behavior should differ from the rest of the distant-water fleet in ways consistent with such motive. We do not assume that every vessel in the fleet engages in surveillance and therefore separate vessels by how strongly each one's presence responds to political distance. For every vessel v, we re-estimate

[27] GDELT relies on automated text processing of news reports rather than manually curated diplomatic records, which makes the data inherently noisy (Hammond and Weidmann, 2014; Hong et al., 2025). Estimates based on this measure are therefore expected to be relatively imprecise.

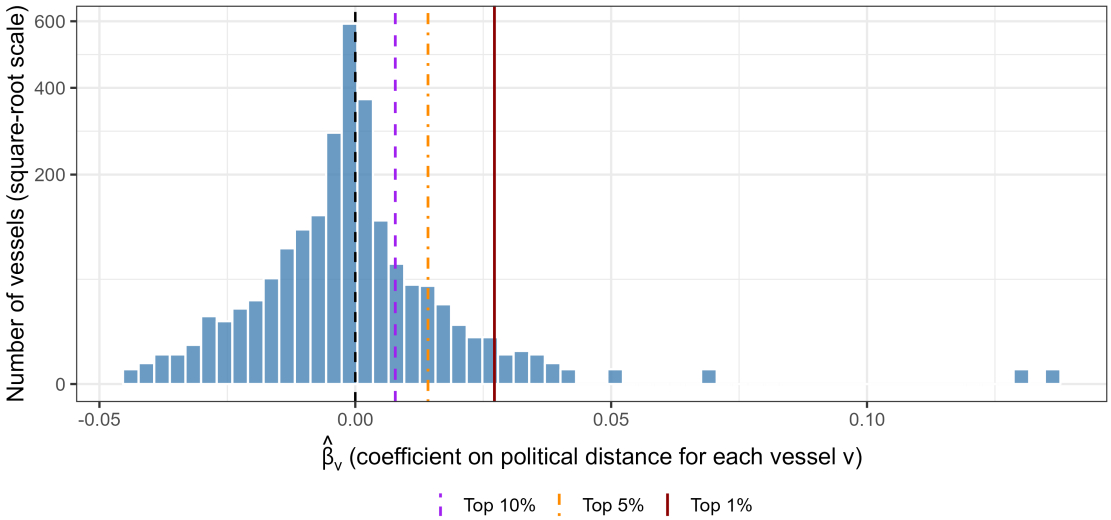


Figure 7: Distribution of the political distance effect across vessels
The figure shows the distribution of 2101 vessel-level OLS coefficients on political distance $\hat{\beta}_v$ from separate regressions of vessel presence on political distance with similar controls as the main regression. Each regression is estimated for one vessel across all EEZ-months in the sample. Vertical lines mark the 90th, 95th, and 99th percentiles of the coefficient distribution. The y-axis is on a square-root scale. The black dashed line indicates a coefficient of zero.

the extensive-margin regression in Eq. 7, yielding 2,101 vessel-specific coefficients $\hat{\beta}_v$:[28]

$$Presence_{v,i,c,t} = \beta_v \, PoliticalDistance_{c,t-24} + \varphi' \mathbf{X}_{i,t} + \alpha_i + \alpha_t + \alpha_{i,m(t)} + \varepsilon_{v,i,c,t}, \qquad \forall v. \quad (8)$$

Figure 7 plots the resulting distribution of vessel-specific coefficients $\hat{\beta}_v$.[29] The distribution has two notable features. First, the majority of vessels have a negative coefficient, consistent with the enforcement channel as these merely fishing vessels appear to be deterred from operating near the EEZs of more politically distant hosts. Second and interestingly, the coefficient distribution is right skewed, with a long tail of vessels whose deployment increases sharply with political distance. This is precisely the pattern we would expect under the surveillance hypothesis if a subset of the fleet conducts intelligence-gathering activity targeted at politically distant hosts.

We proceed to examine whether the observable characteristics and behaviors of vessels with the largest positive $\hat{\beta}_v$ are consistent with surveillance activity.[30] Figure 8 compares the top 10%, 5%, and 1% of vessels (ranked by $\hat{\beta}_v$) against the bottom 90% across the characteristic and behavior measures. We report the full results with raw group means in Appendix

[28] Vessels that are exclusively found in Japanese buffers are discarded as they are likely to be domestic nearshore fishing boats.

[29] The ranking of vessels is based on the political distance coefficients with presence as the outcome, although it is only marginally affected when instead using fishing likelihood (Spearman's rank correlation coefficient is approximately 0.85).

[30] Appendix B.3 and D.2 describe data sources and construction of the vessel-level variable, respectively.

Table D.2. For each characteristic or behavior, we test for mean differences between each top group and the bottom 90% (1,890 vessels) and quantify them in terms of standard deviations of the bottom 90%.

Vessel characteristics. We first look into vessel characteristics. Three patterns emerge from the results in the first panel of Figure 8. First, among the top 1% (22 vessels), none have vessel registries that connect them to state-owned firms. However, as seen in Appendix Table D.2, over 90% of them cannot be connected to any company at all. Second, we observe that nearly all of them broadcast themselves via their AIS to be fishing vessels. These two patterns are suggestive of surveillance motive, as while the former makes these vessels opaque enough to not be connected to the government, the latter makes them inconspicuous so as to minimize the risk of coast guard inspections. Third, when it comes to vessel gear, trawlers are overrepresented in top 1%. Additionally, the top vessels' gear type is more often flagged as ambiguous, and the GFW neural-network classifier (which assigns gear type from vessel characteristics and behavior) is significantly less confident in its assignments for these vessels. Together, these features of untraceable ownership, fishing-vessel identification, and disproportionately trawler-type gear with ambiguous classification largely align with descriptions of the Chinese Maritime Militia that document purpose-built trawlers with communications equipment and military features deployed for intelligence-gathering (Poling et al., 2021).

Behavioral patterns. In addition to the discussed characteristics, we also find anomalies in the behavior of suspected surveillance vessels. Perhaps the most surprising difference is that they disable their automatic identification system (AIS) less. Instead of trying to conceal, it might be possible that that these ships disguise themselves so as to stay as inconspicuous as possible. It is usually vessels that are engaged in illegal fishing in foreign EEZs that turn off AIS in order to conceal their location to local coast guards (e.g., Welch et al., 2022). Therefore, legal compliance by clearly revealing one's location to be in international waters could help minimize the risk of boarding inspections. In addition, consistent with previous results on small fleet sizes, vessels that drive our results share border visits with significantly fewer other Chinese vessels on average. We next look into seasonal patterns of vessel behaviors. If border visits were economically motivated, we would expect them to cluster in fishing seasons. Vessels with stronger political-distance responses yet show less seasonal concentration: their visits are spread more evenly across calendar months than those of vessels in the bottom 90%. Lastly, more politically motivated vessels also seem to spend significantly more of their time in buffer zones right next to the foreign EEZs.

Economic responsiveness. The evidence so far concerns who these vessels are and how

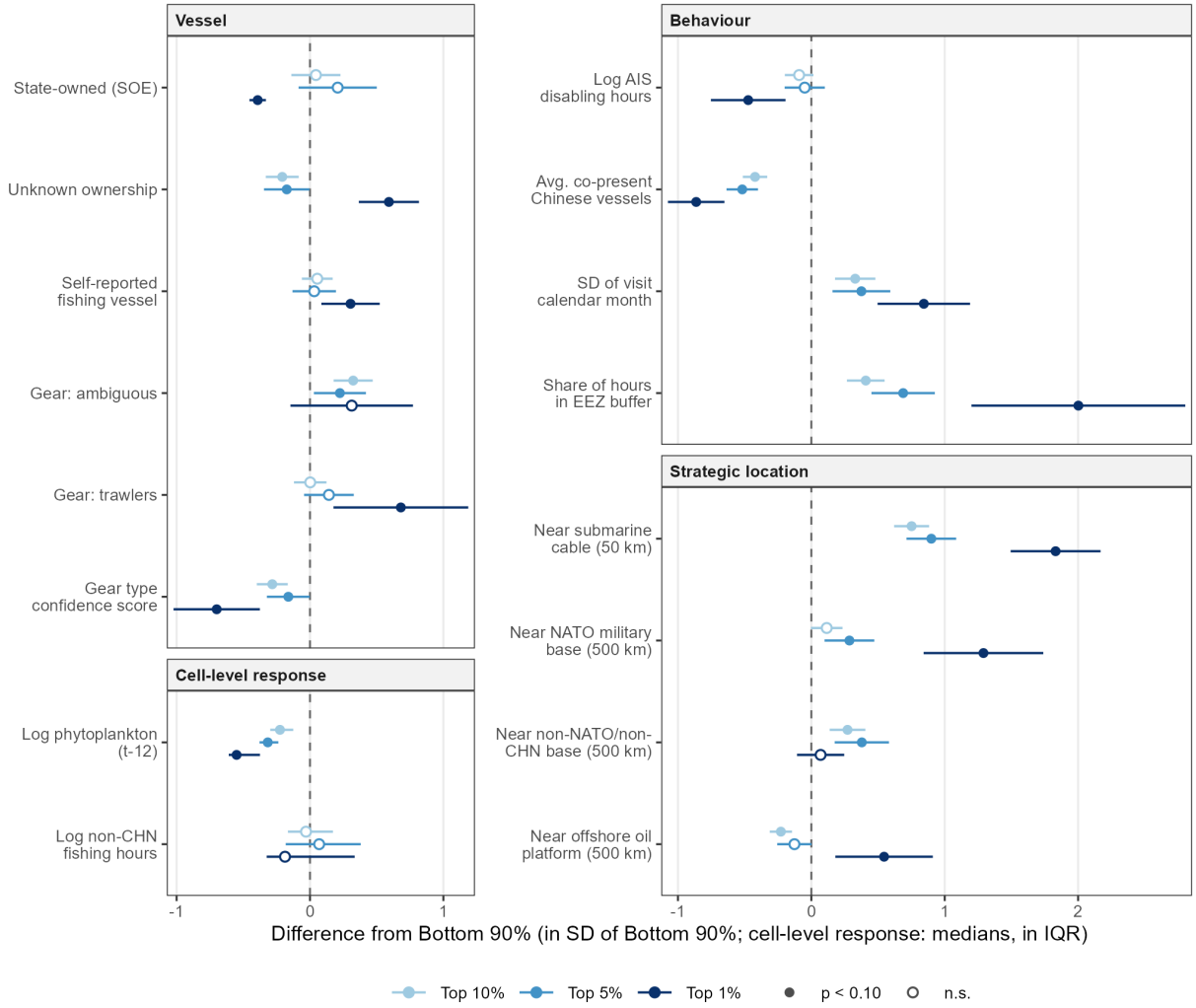


Figure 8: Vessel characteristics by political distance coefficient percentile

Figure shows how the vessels that drive our results differ from the rest of the Chinese fishing fleet. The sample consists of 2,101 Chinese DWF vessels for which a vessel-specific political-distance coefficient $\hat{\beta}_v$ is estimated via equation 8. Vessels are ranked by $\hat{\beta}_v$ and grouped into the top 10%, 5%, and 1% (light to dark blue); the remaining bottom 90% serves as the reference group. Each point shows the difference between the mean of a top group and the mean of the bottom 90%, expressed in standard deviations of the bottom 90%, with a 90% confidence interval from a two-sample t-test of equality of means. The *cell-level response* panel is the exception: its entries are per-vessel PPML coefficients estimated on individual vessel-specific panels and are heavily skewed by outliers, so it shows the difference in group medians, expressed in interquartile ranges of the bottom 90%, with a 90% confidence interval from 2,000 bootstrap resamples of vessels and significance from a Wilcoxon rank-sum test. Filled points indicate a difference significant at the 10% level, hollow points an insignificant one. See Appendix B.3 for data sources, Appendix D.2 for variable definitions and Appendix Table D.2 for results showing variable levels for each group.

they move, but does not yet examine the extent to which they fish for commercial reasons. We test this directly at the level of 0.1° grid cells. For each vessel, we regress with a PPML model its monthly fishing hours in a given cell on our two main economic controls, with cell and month fixed effects, so that the coefficient captures whether a vessel concentrates effort in the months when a cell within its operating area is more favorable for fishing than that cell's own average. Appendix D.2 details the specification. The results show a monotone decline in economic responsiveness for phytoplankton. Among the bottom 90%,

the median vessel raises fishing hours in response to increases in phytoplankton abundance, the pattern expected of a commercially motivated operator. Responsiveness falls steadily across the top percentiles and turns negative for the top 1%, where only 36% of vessels respond positively to phytoplankton at all, against 72% in the bottom 90%. The differences are significant at every threshold. In short, these vessels do not intensify fishing when their own grounds improve. Yet they have not stopped following other fleets. The response to non-Chinese fishing hours in the same cell and month is statistically indistinguishable across all groups, which is consistent with the idea that other nations' fishing vessels provide a salient commercial cover as opposed to environmental conditions.

Strategic locations. Finally, we examine whether potential surveillance vessels target strategically relevant locations, including submarine cables, overseas military bases, and offshore oil rigs. Figure 9 shows a map of these locations. This approach also gives us insight into what type of intelligence is plausibly being collected. In the last panel of Figure 8, we find stark differences in the amount of time vessels spend near submarine cables and near NATO military bases. The more we filter based on the magnitude of the political distance-presence association, the more of the vessel's time is allocated to being in proximity to these strategic locations. Proximity to military installations fits the documented intelligence role of the maritime militia, whose vessels report the movements of foreign warships and approach foreign naval vessels to verify their identity and film their activities (Martinson, 2025). Notably, vessels most responsive to political incentives spend nearly 80% of their time near submarine cable routes, compared with 32% for the remaining fleet (see Appendix Table D.2). This coincides with the over-representation of bottom trawlers among this top group. Bottom trawling is believed to be a leading cause of submarine cable faults (Drew and Hopper, 2009). With accidental breaks this common, sabotage is hard to prove, yet Chinese-linked vessels have been accused of gray-zone operations against cables (Braw, 2023; Hale, 2025). The combination of disproportionate cable-route presence and trawler-heavy gear composition might suggest pre-positioning for sabotage, mapping cable routes or support for fiber tapping. Fishing vessels would then function as instruments of economic statecraft near digital chokepoints over which China has less jurisdiction than western countries (Porcellacchia et al., 2026) and therefore potentially greater benefit of substituting such leverage through interference using fishing vessels.

Taken together, the vessel-level evidence shows that suspected surveillance vessels (i) tend to have untraceable ownership while self-identify as fishing vessels, (ii) are disproportionately equipped with trawler gear consistent with possibly purpose-built Maritime Militia platforms, (iii) respond less to economic incentives, and (iv) concentrate their time near sub-

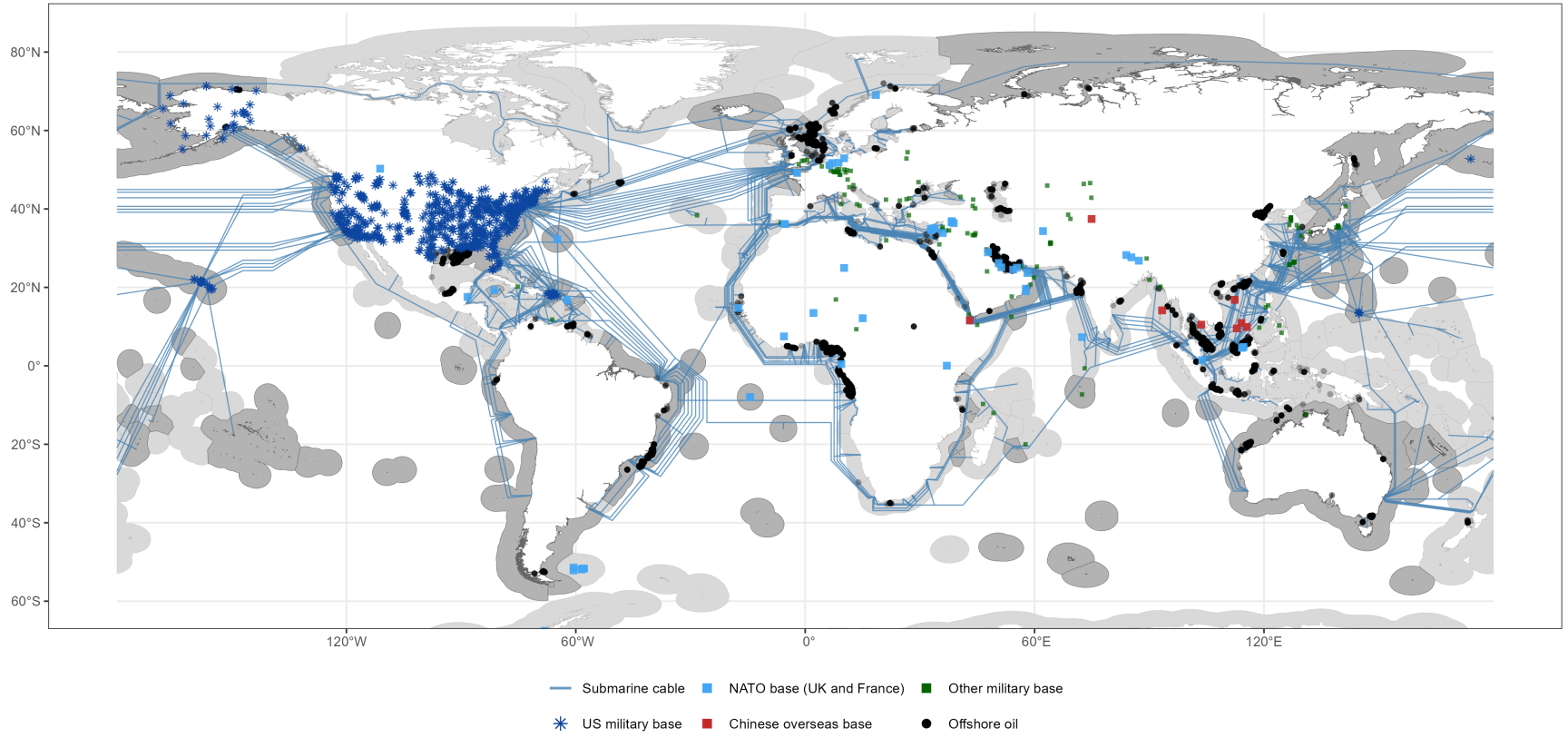


Figure 9: Strategic infrastructure map

Notes: The map shows strategic infrastructure overlaid on EEZs, with sample EEZs in darker grey. US military bases are shown as dark blue asterisks; NATO bases, operated mostly by the UK and France with a few Italian bases, as light blue squares; Chinese overseas bases as red squares; and overseas bases of other countries in dark green. Blue lines show submarine cables active in 2014, and black dots show offshore oil platforms.

marine cables and NATO military installations. These characteristics are not individually decisive but jointly consistent with an underlying profile: inconspicuous, opaque, and likely operationally focused vessels gathering intelligence on critical infrastructure and military assets under commercial cover. This supports surveillance as the primary mechanism behind the political drivers of high seas fishing and establishes DWF as a geoeconomic instrument.

8 Discussion

Our analysis shows that Chinese DWF activity just outside host countries' EEZs rises with political distance from China, with surveillance as the most plausible motive. This has two potential consequences. Politically incentivized fishing may persist where economic returns do not support it, which could worsen resource exploitation. Meanwhile, increased surveillance activity, even when pursued as an ex-ante defensive strategy, can be read as threatening and may thus further strain interstate relations and raise the risk of escalation (Buchanan, 2017; Jervis, 1978). To our knowledge, these mechanisms are not accounted for at three levels. First, standard common-resource models consider only economic incentives and may therefore understate resource degradation. Second, existing international law does not address surveillance incentives specifically. Activities like cable tapping are not illegal on the high seas, while mere suspicion of surveillance does not justify boarding a vessel (United Nations, 1982; Burdette, 2021; Kraska, 2022; Liu and Cui, 2025). Third, international coordinators such as Regional Fisheries Management Organizations (RFMOs) balance economic and conservation interests but have neither the mandate nor the information to weigh geopolitical ones. The channel we document thus falls through the gaps of both the institutions that govern these waters and the research that informs them.

Based on our findings, we discuss measures that (i) host countries can implement in the short run within their jurisdiction and (ii) high-seas governance could pursue as the long-run reform goal. Vessels suspected of surveillance may avoid behaviors targeted by existing measures against illegal fishing, especially if they operate outside foreign EEZs and leave their AIS activated as documented in Section 7. Nevertheless, existing enforcement instruments could still be applied to address these vessels (FAO, 2009). The most direct lever is port access, which DWF vessels rely on for maintenance, crew changes, and resupply (Seto et al., 2023). Here, port states could condition access on continuous tracking and ownership transparency, as any vessel entering a port falls under the port state's jurisdiction (United Nations, 1982). Vessels exhibiting questionable behaviors, such as loitering near submarine cables or military installations, can be inspected upon entry where legally permitted. The

resulting evidence can then help justify restricting or denying port access. Such measures can further be coordinated among coastal states to prevent vessels from moving to neighboring ports (Goodman, 2023). In the long run, international law could establish a common standard, for instance by recognizing persistent presence near critical infrastructure or military installations, as a ground for inspection on the high seas. Because such a global standard requires the agreement of all states, coastal states should work together to build support for it among vessel-sending states through diplomacy (United Nations, 1969).[31] We are careful to note that none of these measures eliminates the underlying motive, which is rooted in geopolitics rather than in the fishery itself. What they can do is make distant-water fishing fleet a less attractive vehicle for such motive, i.e., through pushing surveillance onto costlier platforms that forfeit the civilian cover (Protzer and Stein, 2023). That displacement itself benefits the commons: removing the political payoff from fishing effort would reduce activity to the level justified only by economic reasons.

9 Conclusion

Distant-water fishing fleets on the high seas depend heavily on government subsidies, without which much of their activity would be unprofitable. Such sustained state support raises the question of whether their deployment responds only to economic incentives or also to geopolitical ones. Instruments of economic statecraft typically operate through jurisdiction over the markets, resources, or networks on which a target depends. The high seas, however, are a global commons over which no state holds such jurisdiction, so access can be neither granted nor withdrawn. We show that subsidized fleets can nonetheless serve statecraft in this setting, through sustained presence rather than control over access, with surveillance as the most plausible motivation. Exploiting within-state variation in UNGA voting-based political distance, we find that greater distance between China and a coastal host predicts more Chinese fleet presence and fishing in the adjacent high seas, at both the extensive and intensive margins. Of the mechanisms we consider, the evidence supports surveillance over enforcement, power projection, and retaliation. The effect is concentrated in small fleets near democratic hosts, and the vessels most responsive to political distance are disproportionately of unknown ownership, operate as trawlers, fish less seasonally, respond less to phytoplankton abundance, and cluster near submarine cables and NATO military installations. Unlike the

[31] In this particular case, the political payoffs we document may disincentivize China from agreeing. China has not ratified the UN Fish Stocks Agreement, whose Article 21 lets RFMO members board other parties' vessels on the high seas to enforce conservation measures, and it resists extending RFMO mandates to political matters (United Nations, 1995; Li, 2025). Nonetheless, agreement remains possible, as China has become more cooperative on related issues of illegal fishing and conservation (Li, 2025).

industrial espionage studied in geoeconomics so far, this surveillance thus appears to target military and critical infrastructure. Our extension of the Gordon-Schaefer model shows that such political payoffs can sustain fishing effort beyond economically viable levels. Politically motivated fishing may therefore accelerate stock depletion or contribute to environmental damages. Consequently, governance frameworks focused solely on economic and conservation incentives may prove insufficient to contain such activity.

Three limitations warrant caution in interpreting these results. First, surveillance is inferred from a joint pattern of ownership, gear, timing, economic non-responsiveness, and location rather than observed directly, while direct evidence that countermeasures would require can only come from on-board inspection. Second, our sample excludes areas with overlapping claims, including the South China Sea, where power projection is most likely, and enforcement typically targets unauthorized vessels within EEZs rather than beyond them. Our evidence against both mechanisms therefore applies only to behavior just outside EEZ borders. Third, our design is not quasi-experimental. Although our robustness checks address reverse causation and the most plausible confounders, we remain cautious about claiming causality. On implications for policy, existing frameworks do not reach the vessels we identify, as they comply with international laws and regulations. In the short run, host countries can jointly condition port access on continuous tracking and transparent ownership and inspect suspicious vessels on entry where allowed. In the long run, closing this gap in high-seas governance asks for a common international standard for boarding and inspecting vessels that persistently loiter near critical infrastructure and military installations.

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Appendices

Appendix A Modeling political motivations

We begin with a standard Gordon-Schaefer profit-maximization benchmark built in the work of Gordon (1954); Schaefer (1954, 1957), in which a vessel or sub-fleet i chooses distant-water fishing effort E_i to maximize its economic profit. Let $V_i(\cdot)$ denote the vessel's objective function and let $\Pi_i(E_i, X)$ denote economic profit, where X is the available fish stock.[32] In the purely economic benchmark,

$$V_i(E_i, X) = \Pi_i(E_i, X). \tag{A.1}$$

Following the standard Schaefer harvest specification, harvest is given by

$$H_i = q_i E_i X, \tag{A.2}$$

where q_i is the catchability coefficient. Economic profit is therefore

$$\Pi_i(E_i, X) = \underbrace{p_i\, q_i\, X\, E_i}_{\text{total revenue of fishing}} - \underbrace{(c_i E_i + 0.5\, \gamma_i\, E_i^2)}_{\text{total cost of fishing}}, \tag{A.3}$$

where p_i is the unit price of harvest, c_i is the baseline marginal cost of distant-water fishing effort, and $\gamma_i > 0$ captures increasing marginal costs of effort. The convex cost term reflects the idea that additional effort becomes progressively more costly due to travel distance, congestion, or operational complexity.

The vessel chooses effort to solve

$$\max_{E_i \geq 0} V_i(E_i, X). \tag{A.4}$$

The first-order condition for an interior optimum is

$$\frac{\partial V_i}{\partial E_i} = \frac{\partial \Pi_i}{\partial E_i} = p_i q_i X - c_i - \gamma_i E_i = 0. \tag{A.5}$$

[32] Whereas the canonical Gordon-Schaefer framework treats X as endogenous to aggregate harvest through stock dynamics, we take X as given to the individual vessel's static effort decision. In our setting, a single vessel's harvest is negligible to the total resource available and the empirical analysis examines high-frequency monthly variation.

The second-order condition is satisfied because

$$\frac{\partial^2 V_i}{\partial E_i^2} = -\gamma_i < 0. \tag{A.6}$$

The optimal level of effort under economic profit as the sole objective is

$$E_i^* = \frac{p_i q_i X - c_i}{\gamma_i}. \tag{A.7}$$

We then augment the economic benchmark by allowing effort to generate political payoffs. The vessel's objective becomes

$$V_i(E_i, X) = \Pi_i(E_i, X) + \Phi_i(E_i), \tag{A.8}$$

where $\Phi_i(E_i)$ captures the political payoff from presence and fishing. Based on the possible political constraints or motives discussed in Section 3.1, we model the vessel's decision problem taking α_i^{psr} and α_i^e as parameters reflecting the Chinese state's policy and subsidy environment for its DWF vessels and the host country's enforcement intensity, respectively. A fully structural model would derive these from the underlying interaction between the Chinese state, the vessel operator, and the host state. We treat them as exogenous primitives to generate testable comparative statistics, letting the empirical estimates speak to which factors shape them. We adopt a linear specification for the political payoff:

$$\Phi_i(E_i) = \alpha_i^{psr} E_i - \alpha_i^e E_i, \quad \alpha_i^{psr}, \alpha_i^e \geq 0. \tag{A.9}$$

α_i^{psr} captures power projection, surveillance, or retaliation motives, while α_i^e captures deterrence generated by enforcement from the host country or region. We adopt the linear case to keep the optimal-effort expression interpretable and to focus the empirical analysis on the sign and heterogeneity of net political payoff rather than on identifying the form of Φ.

Define the net political incentive, or political wedge, as

$$\Delta_i \equiv \alpha_i^{psr} - \alpha_i^e. \tag{A.10}$$

The vessel's objective can then be written compactly as

$$V_i(E_i, X) = p_i q_i X E_i - \left(c_i E_i + \frac{\gamma_i}{2} E_i^2\right) + \Delta_i E_i. \tag{A.11}$$

The vessel chooses effort to solve

$$\max_{E_i \geq 0} V_i(E_i, X). \tag{A.12}$$

The first-order condition for an interior optimum is

$$\frac{\partial V_i}{\partial E_i} = p_i q_i X - c_i - \gamma_i E_i + \Delta_i = 0. \tag{A.13}$$

The second-order condition is satisfied because

$$\frac{\partial^2 V_i}{\partial E_i^2} = -\gamma_i < 0. \tag{A.14}$$

Thus, the optimal level of effort under additional political payoff is

$$E_i^{**} = \frac{p_i q_i X - c_i + \Delta_i}{\gamma_i}. \tag{A.15}$$

This yields three empirically relevant scenarios to test how optimal level of vessel's effort changes.

Scenario 1: Neutral political incentives.

If $\alpha_i^{psr} = \alpha_i^e$, then $\Delta_i = 0$ and $E_i^{**} = E_i^*$. This implies that political incentives cancel out, and effort is determined only by economic returns.

Scenario 2: Negative net political incentives.

If $\alpha_i^{psr} < \alpha_i^e$, then $\Delta_i < 0$. Political deterrence from host country lowers the marginal payoff from the vessel's effort:

$$\frac{\partial V_i}{\partial E_i} < \frac{\partial \Pi_i}{\partial E_i}. \tag{A.16}$$

Hence, optimal effort is lower than in the purely economic benchmark:

$$E_i^{**} < E_i^*. \tag{A.17}$$

In this case, fishing may cease even when the economic marginal return is still positive, as long as the total marginal payoff is non-positive:

$$\frac{\partial \Pi_i}{\partial E_i} > 0 \quad \text{and} \quad \frac{\partial \Pi_i}{\partial E_i} + \Delta_i \leq 0. \tag{A.18}$$

Thus, enforcement can deter fishing before the vessel reaches its profit-maximizing capacity.

Scenario 3: Positive net political incentives.

If $\alpha_i^{psr} > \alpha_i^e$, then $\Delta_i > 0$. Political motives increase the marginal payoff from effort:

$$\frac{\partial V_i}{\partial E_i} > \frac{\partial \Pi_i}{\partial E_i}. \tag{A.19}$$

Hence, optimal effort is higher than in the purely economic benchmark:

$$E_i^{**} > E_i^*. \tag{A.20}$$

In this case, fishing may continue even when the economic marginal return is negative, as long as the total marginal payoff remains positive:

$$\frac{\partial \Pi_i}{\partial E_i} < 0 \quad \text{and} \quad \frac{\partial \Pi_i}{\partial E_i} + \Delta_i > 0. \tag{A.21}$$

Thus, politically motivated effort can persist even under economic losses.

In Sections 4 to 7, we test the hypotheses depicted in these three scenarios empirically.

Appendix B Data

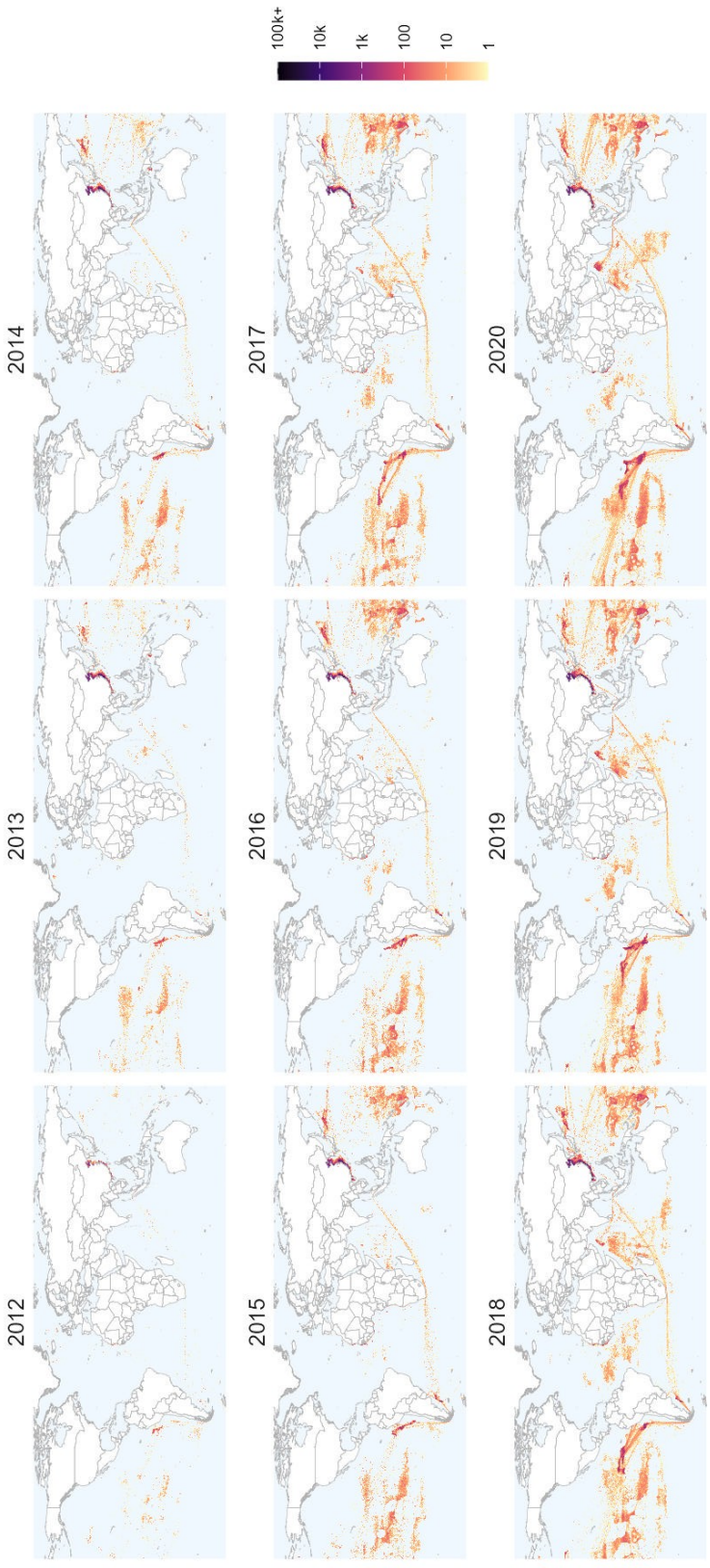


Figure B.1: Extent of the Chinese DWF fleet presence over time
Figure plots the annual total presence hours of the Chinese DWF fleet at the 0.1° grid-cell level across the global ocean from 2012 to 2020. Presence hours are shown on a log scale, ranging from 1 hour (lightest) to over 100,000 hours (darkest). The map shows an expansion of the Chinese DWF fleet's spatial reach over the sample period, with concentrated activity in the Southeast Pacific (off Argentina, Chile, Peru, and Ecuador), Northwest Pacific (around Japan and Russia), and West African coast. Vessel position data are obtained from Global Fishing Watch.

Appendix B.1 Summary statistics of the main data

Table B.1: List of exclusive economic zones

	Host country	EEZ name
1	Argentina	Argentinean Exclusive Economic Zone
2	Australia	Australian Exclusive Economic Zone
3	Australia	Australian Exclusive Economic Zone (Macquarie Island)
4	Australia	Christmas Island Exclusive Economic Zone
5	Australia	Cocos Islands Exclusive Economic Zone
6	Australia	Heard and McDonald Islands Exclusive Economic Zone
7	Australia	Norfolk Island Exclusive Economic Zone
8	Brazil	Brazilian Exclusive Economic Zone
9	Brazil	Brazilian Exclusive Economic Zone (Trindade)
10	Chile	Chilean Exclusive Economic Zone
11	Chile	Chilean Exclusive Economic Zone (Easter Island)
12	Chile	Chilean Exclusive Economic Zone (San Felix and San Ambrosio islands)
13	Comores	Comoran Exclusive Economic Zone
14	Ecuador	Ecuadorian Exclusive Economic Zone
15	Ecuador	Ecuadorian Exclusive Economic Zone (Galapagos)
16	France	Amsterdam Island & St. Paul Island Exclusive Economic Zone
17	France	Bassas da India Exclusive Economic Zone
18	France	Clipperton Exclusive Economic Zone
19	France	Crozet Islands Exclusive Economic Zone
20	France	French Exclusive Economic Zone
21	France	French Guiana Exclusive Economic Zone
22	France	French Polynesian Exclusive Economic Zone
23	France	Guadeloupean Exclusive Economic Zone
24	France	Ile Europa Exclusive Economic Zone
25	France	Juan de Nova Exclusive Economic Zone
26	France	Kerguelen Exclusive Economic Zone
27	France	Martinican Exclusive Economic Zone
28	France	New Caledonian Exclusive Economic Zone
29	France	Réunion Exclusive Economic Zone
30	France	Saint-Barthélemy Exclusive Economic Zone
31	France	Saint-Martin Exclusive Economic Zone
32	France	Saint-Pierre and Miquelon Exclusive Economic Zone
33	France	Wallis and Futuna Exclusive Economic Zone
34	Iceland	Icelandic Exclusive Economic Zone
35	Japan	Japanese Exclusive Economic Zone
36	Kenya	Kenyan Exclusive Economic Zone
37	Mauritius	Chagos Archipelago Exclusive Economic Zone
38	Nigeria	Nigerian Exclusive Economic Zone
39	Oman	Omani Exclusive Economic Zone

40	Peru	Peruvian Exclusive Economic Zone
41	Russia	Russian Exclusive economic Zone
42	Senegal	Senegalese Exclusive Economic Zone
43	Tonga	Tongan Exclusive Economic Zone
44	United Arab Emirates	United Arab Emirates Exclusive Economic Zone
45	United Kingdom	Anguilla Exclusive Economic Zone
46	United Kingdom	Ascension Exclusive Economic Zone
47	United Kingdom	Bermudian Exclusive Economic Zone
48	United Kingdom	British Virgin Islands Exclusive Economic Zone
49	United Kingdom	Cayman Islands Exclusive Economic Zone
50	United Kingdom	Guernsey Exclusive Economic Zone
51	United Kingdom	Jersey Exclusive Economic Zone
52	United Kingdom	Montserrat Exclusive Economic Zone
53	United Kingdom	Pitcairn Islands Exclusive Economic Zone
54	United Kingdom	St. Helena Exclusive Economic Zone
55	United Kingdom	Tristan Da Cunha Exclusive Economic Zone
56	United Kingdom	Turks and Caicos Exclusive Economic Zone
57	United Kingdom	United Kingdom Exclusive Economic Zone
58	United States	American Samoa Exclusive Economic Zone
59	United States	Guam Exclusive Economic Zone
60	United States	Howland and Baker Islands Exclusive Economic Zone
61	United States	Jarvis Island Exclusive Economic Zone
62	United States	Johnston Atoll Exclusive Economic Zone
63	United States	Northern Mariana Exclusive Economic Zone
64	United States	Palmyra Atoll Exclusive Economic Zone
65	United States	Puerto Rican Exclusive Economic Zone
66	United States	United States Exclusive Economic Zone
67	United States	United States Exclusive Economic Zone (Alaska)
68	United States	United States Exclusive Economic Zone (Hawaii)
69	United States	Virgin Islander Exclusive Economic Zone
70	United States	Wake Island Exclusive Economic Zone
71	Yemen	Yemeni Exclusive Economic Zone

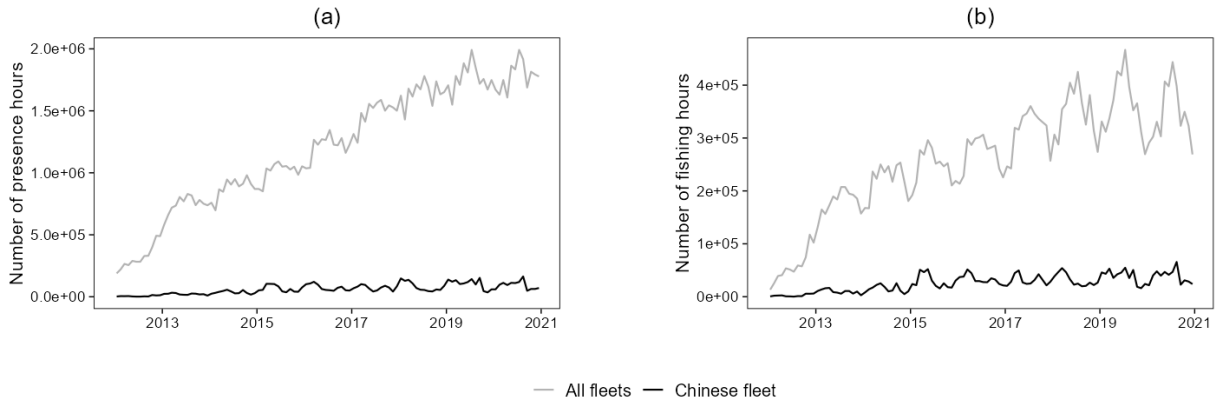


Figure B.2: DWF fleet presence and fishing effort over time
Panel (a) shows the monthly total number of vessel presence days for all fleets (gray line) and for Chinese fleet (black line) recorded for the 50 km buffers of analyzed EEZs from 2012 to 2021. Panel (b) shows the monthly total number of vessel fishing hours as inferred by Global Fishing Watch for all fleets (gray line) and for Chinese fleet (black line) recorded for the 50 km buffers of analyzed EEZ's from 2012 to 2021.

Table B.2: Summary statistics

Variable	Observations	Mean	Std. Dev.	Min.	Max.
Chinese fleet presence	7,668	0.368	0.482	0	1
Chinese fleet fishing	7,668	0.238	0.426	0	1
# presence hours Chinese fleet	7,668	873	6,250	0	137,174
# presence hours all fleets	7,668	16,960	50,068	0	479,548
# fishing hours Chinese fleet	7,668	359	2,322	0	45,051
# fishing hours all fleets	7,668	3,669	12,215	0	128,026
# unique vessels Chinese fleet	7,668	7.471	27.213	0	369
Political distance (Δ ideal-point)	7,668	1.825	1.003	0	3.494
Phytoplankton abundance	7,668	192.930	424.578	1.000	5,554.548
Sea-route distance (km)	7,668	13,764	5,443	2,132	25,522

Appendix B.2 GDELT event-based conflict severity measure

The Global Database of Events, Language, and Tone (GDELT) monitors news media worldwide and automatically codes reported interactions between countries into discrete, geolocated, and time-stamped events (Leetaru and Schrodt, 2013). Following Davis et al. (2019), we construct a measure of the cumulative severity of negative events between China and host countries, restricting the sample to events in which both actors are governments or militaries. Each event carries a Goldstein score ranging from -10 to 10 that quantifies its hypothesized impact on bilateral relations: cooperative acts such as extending military or economic aid score near the top of the scale, whereas hostile acts such as armed attacks score near the bottom.

$$\text{DailySeverity}_{c,d} = \max\Big(\big\{-G_e : e \in \mathcal{E}_{c,d},\ G_e < 0\big\} \cup \{0\}\Big) \tag{B.1}$$

$$\text{Severity}_{c,t}^{(k)} = \sum_{\tau=t-k}^{t-1} \sum_{d \in \mathcal{D}(\tau)} \text{DailySeverity}_{c,d}, \qquad k \in \{4, 12\} \tag{B.2}$$

where $\mathcal{E}_{c,d}$ denotes the set of GDELT events between host country c and China on day d, $G_e \in [-10, 10]$ is the Goldstein score of event e, and $\mathcal{D}(\tau)$ is the set of days in month τ. Equation (B.1) retains only the most severe negative event per country-day, avoiding double-counting of events reported by multiple media outlets (Hong et al., 2025); days without negative events contribute zero. We focus on conflictual interactions by retaining only events with negative scores and multiplying them by -1, so that larger values reflect greater severity. Equation (B.2) aggregates daily severities into a rolling sum over the k months preceding month t. The short-term measure ($k = 4$) is designed to examine potential retaliatory reactions by China; we therefore restrict $\mathcal{E}_{c,d}$ to events in which the host country is the subject and China the object. The 12-months rolling sum, on the other hand, is a broader, slower moving indicator that aims to serve as an alternative main independent variable.

As GDELT codes events automatically from news reports, it offers far broader and more timely coverage than manually curated event datasets, at the cost of noise: validation studies document miscoded event fields and duplicate records (Hong et al., 2025), as well as weaker correspondence with hand-coded data in fine-grained subnational applications (Hammond and Weidmann, 2014). We mitigate these issues by retaining only the most severe event per country-day and aggregating to the country level. Nevertheless, we expect estimates based on this measure to be less precise due to remaining measurement error.

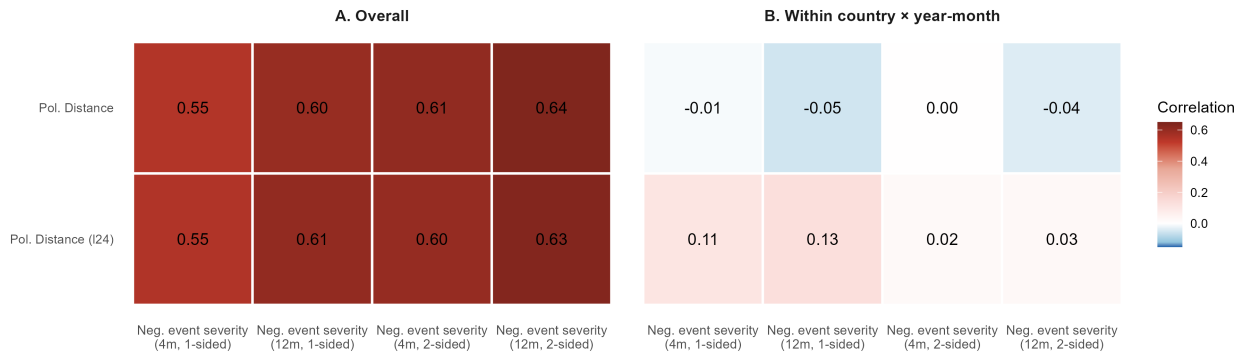


Figure B.3: Correlation between political distance and GDELT negative-event severity
Panel (A) shows the raw (overall) Pearson correlations between two measures of UN ideal-point political distance from China, contemporaneous (*Pol. Distance*) and 24-month lagged (*Pol. Distance (l24)*). The four GDELT-based negative-event severity measures are varying the rolling window (4 versus 12 months) and directionality (one-sided: events directed at China by the host country; two-sided: bilateral events in both directions). Panel (B) shows the same correlations after two-way demeaning by country and year-month fixed effects, isolating within-country temporal co-movement. The main sample is used to calculate the corelation matrices.

Figure B.3 validates the GDELT-based severity measures against the ideal-point distance used in the main analysis. Panel A shows large positive raw correlations between *Pol. Distance* and all four event-severity measures (0.55–0.64): countries at greater geopolitical distance from China indeed experience more negative bilateral events. Panel B isolates within-country temporal co-movement by demeaning both variables with country and year-month fixed effects. The correlations with lagged political distance, that takes the temporal structure into account, remain positive but drop considerably (0.02–0.13): a country moving away from China's UN voting position sees more negative bilateral events, yet most of the raw correlation reflects cross-country differences. This divergence is by design. Ideal-point distances capture slow-moving ideological alignment, whereas the GDELT measure captures discrete diplomatic transgressions; their limited within-country co-movement confirms that the severity measure provides variation distinct from political distance rather than merely proxying for it.

Appendix B.3 Additional data

Shapefiles and sea-route distance

EEZ boundaries. For buffer construction, we use the Exclusive Economic Zone boundary shapefiles from the World EEZ data version 11 provided by Flanders Marine Institute (2019).

Sea routes. To calculate sea-route distances between China's EEZ and other EEZs in the sample, we draw on the `searoute` Python package developed by Halili (2024).

Additional control variables

Sea surface temperature. For the additional controls reported in Appendix Table C.1, we use NOAA Optimum Interpolation Sea Surface Temperature (OISST) records developed by Huang et al. (2021). Raw daily OISST data at the 0.25° grid-cell level are aggregated to construct mean sea surface temperature and sea surface temperature anomalies relative to the long-run climatic average at the buffer-month level.

Storms. We calculate the number of cyclones passing through each buffer using data from the International Best Track Archive for Climate Stewardship (IBTrACS) Project version 4.01 developed by Knapp et al. (2010); Gahtan et al. (2024). We aggregate the raw 3-hour interval data at the 0.1° grid-cell level to count the number of unique storms observed within a given buffer-month.

Economic variables. We additionally incorporate total bilateral trade volume between China and the host country, as reported by the IMF, together with an indicator for the presence of a regional trade agreement documented by the WTO. We also control for host-country income using GDP per capita. All economic variables are drawn from the CEPII Gravity database developed by Conte et al. (2022).

Aid and trade dependency. Chinese aid is added from AidData (2021) and divided by GDP (World Bank, 2024) for the aid dependency measure. Trade dependence is calculated analogously with data from Conte et al. (2022) for both bilateral trade volume and GDP.

Alternative political distance measures

Vote disagreement. This measure is based on the complete vote from the same source as the main variable (Bailey et al., 2017). It is a linear transformation of commonly used agreement scores that flips the sign of the coefficient so that it is coherent with political distance.

UNGA DM. These are the ideal point distances calculated based on an extended UNGA voting data set created by Fjelstul et al. (2023, 2025). It contains a comprehensive database of all resolution-related decisions taken in UNGA plenary sessions from 1946 to the present,

covering not only the roll-call votes on adopted resolutions captured by existing datasets but also votes on failed resolutions, amendments, motions, and separate votes, as well as consensus and non-recorded decisions. The measure for important votes calculates ideal point distances based exclusively on the votes the US State Department deemed as "important".

Instrumental variable construction

We apply the UNGA vote issues classified by Voeten (2013). The observed country votes with issue tags come from datasets `un_votes`, `un_roll_calls`, and `un_roll_call_issues` in the `unvotes` R package developed by Robinson (2021).

Data for mechanism analysis

Vessel ownership. Individual vessels were linked to companies by Carmine et al. (2020), who utilize public databases such as vessel registries. We then manually looked up Chinese companies at Qichacha (QCC) (2026), one of the most commonly used Chinese business registry to identify which firms are private and which are state-owned.

Vessel characteristics. All vessel characteristics are inferred from Global Fishing Watch (2024), with the methodology described in Kroodsma et al. (2018). Gear is estimated based on behavior.

AIS disabling. Welch et al. (2022) provide AIS disabling data that distinguishes between accidental disabling events from those likely done with intention.

Strategic locations. We adopt submarine cables from TeleGeography (2026); with the resulting data being cables that where active in 2014. For military bases, we use data from Man and Palmer (2022) for overseas bases of China, France, India, Japan, Italy, Russia, United Arab Emirates, United Kingdom and the US. We supplement US bases with data from U.S. Department of Transportation (2026) for completeness. Lastly, for offshore oil bases, we use satellite-based estimates from World Bank Global Flaring and Methane Reduction Partnership et al. (2025) and filter for offshore platforms.

Appendix C Validity and robustness tests

Appendix C.1 Examining reverse causality

To assess the presence of reverse causality between DWF fleet activity and interstate political distance, we estimate the following dynamic model:

$$y_{i,c,t} = \sum_{k \in \mathcal{K}} \beta_k, PoliticalDistance_{c,t-k} + \varphi' \mathbf{X}_{i,t} + \alpha_i + \alpha_t + \alpha_{i,m(t)} + \varepsilon_{i,c,t}, \tag{C.1}$$

where $\mathcal{K} = \{-24, -12, 0, 12, 24\}$. Positive values of k correspond to lags of political distance, allowing us to test whether past political distance affects current fleet activity—the causal direction. Negative values correspond to leads of political distance, allowing us to test whether current fleet activity predicts contemporaneous or future political distance—the reverse-causality direction. The outcome $y_{i,c,t}$ is the probability of the Chinese DWF fleet present or fishing in the 50 km buffer just outside a given EEZ. In analogy to the estimation in Eq. 7, we factor in the EEZ, time, and seasonality fixed effects and time-varying controls.

Figure C.1 plots the resulting coefficients. In both panels, the pattern is consistent with a causal interpretation. In Panel (a) for outcome measuring the probability of fleet presence: at positive lags (12 and 24 months), the coefficient on political distance is positive and statistically significant at the 24-month horizon, which matches our headline specification reported in Table 1. At lead horizons (-12 and -24 months), the coefficient is statistically indistinguishable from zero at the 12-month lead but turns *negative* at the 24-month lead, with the point estimate marginally significant. The negative sign is the opposite of what reverse causality would predict if Chinese fleet presence were aggravating bilateral relations—which would result in a positive lead coefficient. Nonetheless, the contemporaneous and 12-month-lead coefficients are null. Panel (b) shows the same pattern for the probability of fleet fishing, with significance yet concentrated only at the 24-month lag. Overall, we intepret the temporal asymmetry—significant positive coefficients at lags but not at leads—to be consistent with political distance driving fleet activity rather than the reverse.

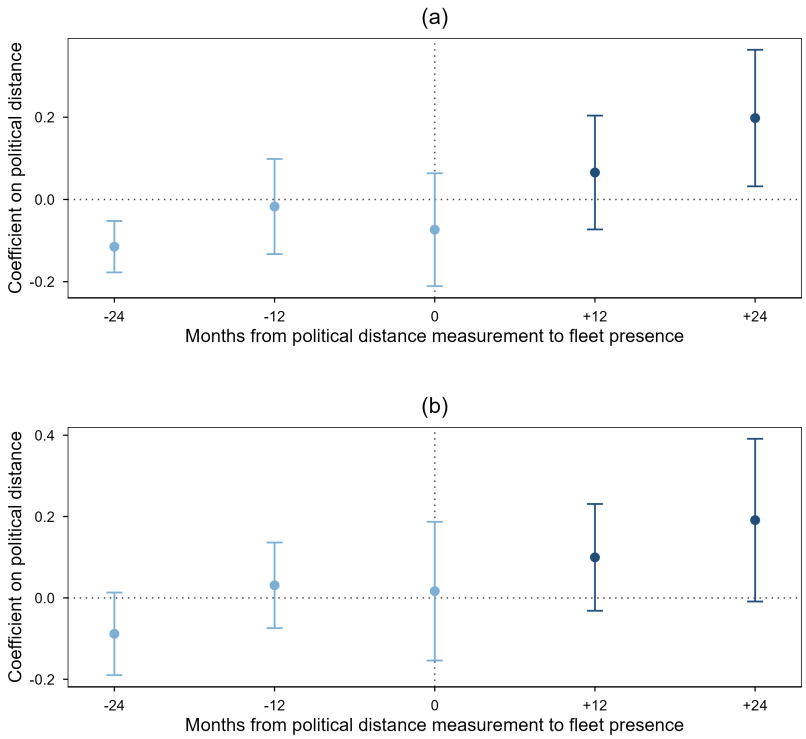


Figure C.1: Dynamic coefficients on political distance

Figure plots coefficients from an the specification regressing the probability of Chinese DWF fleet presence (Panel a) and fishing (Panel b) in the buffer just outside the host country's EEZ on leads and lags of political distance between China and the host country (Equation C.1). The x-axis represents months between political distance measurement and outcome. Positive values (lags) test whether past political distance affects current fleet activity (causal direction); negative values (leads) test whether current fleet activity predicts future political distance (reverse-causality direction). Vertical bars denote 95% confidence intervals based on standard errors clustered at the country level.

Table C.1: Inclusion of further ecological and economic control variables

	1(CHN fleet presence)					1(CHN fleet fishing)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Political distance $_{t-24}$	0.149**	0.147**	0.140*	0.146**	0.136*	0.167*	0.165*	0.158	0.160*	0.153*
	(0.072)	(0.072)	(0.072)	(0.068)	(0.068)	(0.091)	(0.091)	(0.090)	(0.083)	(0.083)
	[0.045]	[0.050]	[0.068]	[0.037]	[0.065]	[0.084]	[0.087]	[0.101]	[0.073]	[0.095]
Log non-CHN fishing hours	0.026***	0.026***	0.026***	0.026***	0.026***	0.024***	0.024***	0.025***	0.025***	0.025***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
	[0.000]	[0.000]	[0.001]	[0.000]	[0.001]	[0.001]	[0.001]	[0.001]	[0.000]	[0.001]
Log phytoplankton $_{t-12}$	0.035	0.039*	0.039*	0.032	0.039**	0.022	0.025	0.025	0.017	0.022
	(0.020)	(0.018)	(0.019)	(0.018)	(0.015)	(0.023)	(0.021)	(0.022)	(0.021)	(0.019)
	[0.117]	[0.060]	[0.071]	[0.115]	[0.030]	[0.372]	[0.278]	[0.311]	[0.455]	[0.275]
Sea surface temp. anomaly $_{t-12}$		0.032			0.031		0.023			0.023
		(0.017)			(0.017)		(0.020)			(0.020)
		[0.182]			[0.210]		[0.413]			[0.435]
Number of storms		0.020			0.019		0.015			0.015
		(0.013)			(0.013)		(0.008)			(0.008)
		[0.415]			[0.405]		[0.145]			[0.148]
Log total trade $_{t-24}$			-0.014		0.038			-0.063		0.010
			(0.045)		(0.054)			(0.037)		(0.041)
			[0.766]		[0.527]			[0.262]		[0.821]
Has trade agreement $_{t-24}$			-0.089		-0.099			-0.043		-0.061
			(0.051)		(0.043)			(0.044)		(0.035)
			[0.191]		[0.235]			[0.314]		[0.240]
Log host GDP p.c. $_{t-24}$				-0.128	-0.182				-0.242	-0.263
				(0.149)	(0.165)				(0.137)	(0.149)
				[0.446]	[0.346]				[0.112]	[0.123]
EEZ FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Month-year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
EEZ-month FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Mean dependent variable	0.368	0.368	0.368	0.368	0.368	0.238	0.238	0.238	0.238	0.238
Observations	7,668	7,668	7,668	7,668	7,668	7,668	7,668	7,668	7,668	7,668

Notes: 'CHN fleet presence' is an indicator variable equal to one if any presence hour is recorded for the Chinese fleet in the buffer, and zero otherwise. 'CHN fishing presence' is an indicator variable equal to one if any fishing hour is recorded for the Chinese fleet in the buffer, and zero otherwise. 'Political distance' is the absolute difference in UNGA voting-based ideal point between China and the EEZ's host country two year prior. 'Log sea-route distance' is the logarithmized sea-route distance from the centroid of China's EEZ to the centroid of the host EEZ. 'Log non-CHN fishing hours' is the logarithmized number of fishing hours recorded for all non-Chinese fleets in the buffer. 'Log phytoplankton' is the logarithmized amount of phytoplankton abundance measured for the buffer one year prior. 'Sea temp. anomaly' is the deviation of average sea surface temperature in the buffer from its climatological mean one year prior, measured in degrees Celsius. 'Number of storms' is the count of tropical cyclones passing through the buffer in a given month. 'Log total trade' is the logarithmized total bilateral trade volume between China and the host country two years prior. 'Has trade agreement' is an indicator equal to one if China and the host country have any regional trade agreement in force two years prior, and zero otherwise. 'Log host GDP p.c.' is the logarithmized GDP per capita of the host country two years prior. Standard errors clustered at the country level are reported in the parentheses. Wild cluster bootstrap p-values are in square brackets, clustered at the country level. *p<0.1; **p<0.05; ***p<0.01, based on wild cluster bootstrap p-values.

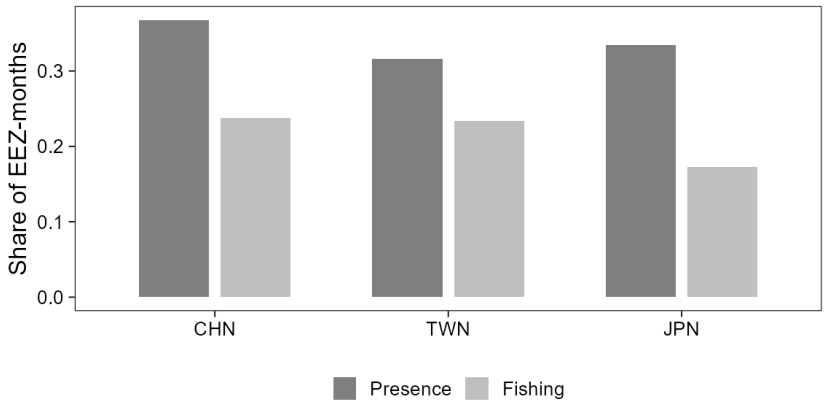


Figure C.2: Chinese, Taiwanese, and Japanese DWF fleet activity
Figure compares the probabilities of observing Chinese, Taiwanese, and Japanese distant-water fishing fleets being present or actively fishing within the 50km buffer outside the EEZs in our sample over the period 2012-2020.

Table C.2: Placebo tests with Taiwanese and Japanese DWF fleet activity as outcomes

	1(TWN presence)		1(TWN fishing)		1(JPN presence)		1(JPN fishing)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Political distance $_{t-24}$	0.030	0.021	0.017	-0.002	0.012	0.011	-0.015	-0.019
	(0.049)	(0.049)	(0.052)	(0.051)	(0.032)	(0.031)	(0.031)	(0.033)
	[0.552]	[0.687]	[0.758]	[0.957]	[0.744]	[0.751]	[0.660]	[0.583]
Log (non-TWN fishing hours)		0.014***		0.024***				
		(0.004)		(0.005)				
		[0.005]		[0.001]				
Log (non-JPN fishing hours)						0.001		0.003
						(0.003)		(0.007)
						[0.711]		[0.843]
Log phytoplankton $_{t-12}$		-0.036*		-0.039**		-0.002		0.005
		(0.022)		(0.017)		(0.014)		(0.015)
		[0.095]		[0.044]		[0.892]		[0.756]
EEZ FE	✓	✓	✓	✓	✓	✓	✓	✓
Month-year FE	✓	✓	✓	✓	✓	✓	✓	✓
EEZ-month FE	–	✓	–	✓	–	✓	–	✓
Mean dependent variable	0.316	0.316	0.234	0.234	0.324	0.324	0.161	0.161
Observations	7,668	7,668	7,668	7,668	7,560	7,560	7,560	7,560

Notes: Tables presents estimates of the placebo tests examining the association between (i) the political distance between China and host country and (ii) activity of DWF fleets similar to China's in terms of economic incentives. 'TWN presence' is an indicator variable equal to one if any presence hour is recorded for the Taiwanese fleet in the buffer, and zero otherwise. 'TWN fishing presence' is an indicator variable equal to one if any fishing hour is recorded for the Taiwanese fleet in the buffer, and zero otherwise. 'JPN presence' is an indicator variable equal to one if any presence hour is recorded for the Japanese fleet in the buffer, and zero otherwise. 'JPN fishing presence' is an indicator variable equal to one if any fishing hour is recorded for the Japanese fleet in the buffer, and zero otherwise. 'Political distance' is the absolute difference in UNGA voting-based ideal point between *China* and the EEZ's host country two years prior. 'Log non-TWN fishing hours' and 'Log non-JPN fishing hours' are, respectively, the logarithmized number of fishing hours recorded for all non-Taiwanese and non-Japanese fleets in the buffer. 'Log phytoplankton' is the logarithmized amount of phytoplankton abundance measured for the buffer one year prior. Columns (5)-(8) excludes observations pertaining to the Japanese EEZs in the sample. Standard errors clustered at the country level are reported in the parentheses. Wild cluster bootstrap p-values are in square brackets, clustered at the country level. *p<0.1; **p<0.05; ***p<0.01, based on wild cluster bootstrap p-values.

Appendix C.3 Measuring political distance

Table C.3: Alternative political distance measures

	1(CHN fleet presence)				
	(1)	(2)	(3)	(4)	(5)
Political distance (baseline)	0.149**				
	(0.072)				
	[0.045]				
Vote disagreement		0.406*			
		(0.193)			
		[0.068]			
Political distance (UNGA-DM)			-0.045		
			(0.114)		
			[0.717]		
Political distance (UNGA-DM, imp. votes)				0.376***	
				(0.089)	
				[0.006]	
Neg. bilateral events (GDELT)					0.042
					(0.022)
					[0.742]
Observations	7,668	7,668	7,668	7,668	7,668
Controls	✓	✓	✓	✓	✓
EEZ FE	✓	✓	✓	✓	✓
Month-year FE	✓	✓	✓	✓	✓
EEZ × month FE	✓	✓	✓	✓	✓

Notes: Columns vary the political distance or bilateral conflict measure. Vote disagreement is inverse of agreement (1 = no agreement, 0 = perfect alignment). UNGA-DM ideal point distances are from Fjelstul et al. (2023, 2025). Negative bilateral event severity is the standardized cumulative Goldstein-scaled severity of adverse bilateral events between China and host countries over the preceding 12 months (two-sided). Controls include log non-CHN fishing hours and log phytoplankton abundance (12-month lag). Standard errors clustered at the country level in parentheses. Wild cluster bootstrap p-values in brackets, clustered at the country level (Rademacher weights, 9,999 replications, null hypothesis imposed). *p<0.1; **p<0.05; ***p<0.01: based on the wild cluster bootstrap p-values.

Table C.4: Vote buying and economic dependency on China

	1(CHN fleet presence)				
	(1)	(2)	(3)	(4)	(5)
Political distance$_{t-24}$	0.149**	0.146*	0.317***	0.347***	0.220*
	(0.072)	(0.074)	(0.045)	(0.071)	(0.086)
	[0.045]	[0.063]	[0.010]	[0.009]	[0.065]
Log non-CHN fishing hours	0.026***	0.026***	0.022**	0.021***	0.021**
	(0.003)	(0.003)	(0.002)	(0.002)	(0.001)
	[0.000]	[0.000]	[0.029]	[0.006]	[0.017]
Log phytoplankton$_{t-12}$	0.035	0.035	0.007	0.008	-0.018
	(0.020)	(0.020)	(0.035)	(0.033)	(0.041)
	[0.117]	[0.117]	[0.853]	[0.829]	[0.699]
CHN aid/GDP$_{t-24}$		1.304			
		(0.921)			
		[0.233]			
CHN trade/GDP$_{t-24}$		0.347			
		(0.892)			
		[0.717]			
Controls	✓	✓	✓	✓	✓
EEZ FE	✓	✓	✓	✓	✓
Month-year FE	✓	✓	✓	✓	✓
EEZ × month FE	✓	✓	✓	✓	✓
Sample	Full	Full	Low aid + trade dep.	Least + low susc.	Least susc.
Observations	7,668	7,668	5,184	6,480	5,508
Countries	21	21	6	11	5

Note: Col. (1): baseline specification. Col. (2): additionally controls for Aid/GDP and Trade/GDP, where Aid/GDP is the ratio of Chinese official finance commitments (AidData CLG 1.0) to host GDP (World Bank), and Trade/GDP is bilateral trade with China as a share of host GDP (CEPII Gravity/BACI), both lagged 24 months. Col. (3): sample restricted to countries with below-median average Aid/GDP and below-median average Trade/GDP (intersection). Col. (4): sample restricted to countries with low susceptibility to Chinese diplomatic pressure due to their status as regional power (US, UK, France, Japan, Australia, Brazil, Russia, Chile, Argentina, Peru, UAE). Col. (5): sample restricted to countries least susceptible to Chinese diplomatic pressure due to their status as major western-aligned power (US, UK, France, Japan, Australia). Controls include log non-CHN fishing hours and log phytoplankton abundance (12-month lag). Standard errors clustered at the country level in parentheses. Wild cluster bootstrap p-values in brackets, clustered at the country level (Rademacher weights, or Webb weights in samples with fewer than 12 countries, 9,999 replications, null hypothesis imposed). *p<0.1; **p<0.05; ***p<0.01, based on the wild cluster bootstrap p-values.

Appendix C.4 Two-stage least squares estimation

A central threat to identification in Eq. 7 is that political distance between a host country and China may be correlated with unobserved determinants of Chinese DWF fleet activity just outside the host's exclusive economic zone. Results discussed in Section C.1 and C.2 suggest that our estimates are unlikely to be driven by reverse causality or by omitted ecological and economic confounders. Two concerns remain. First, since the political-distance measure is based on all UNGA votes, it may capture economic ties shaping fishing activities that we may not observe. Second, the ideal points may be measured with error, which would bias the OLS estimates. We address these concerns by estimating Eq. 7 with two-stage least squares. Specifically, we instrument the baseline political distance with China-host vote disagreement on specific issues which best characterize geopolitical divergence and do not reflect fisheries-relevant economic or regulatory differences. Among the UNGA vote issues classified by Voeten (2013), we use votes on (1) nuclear weapons and nuclear materials proliferation and (2) arms control and disarmament.[33] These issues target security-related topics and have no direct connection with fishing access, trade, or maritime regulation. For each issue, we construct an instrument as the share of issue-specific roll calls on which the host country and China voted differently, following the agreement-score tradition of Signorino and Ritter (1999).[34] The IV is based on the following construction.

$$disagreement^{I}_{c,t} = \frac{1}{|R^{I}_{c,t}|} \sum_{r \in R^{I}_{c,t}} 1[v_{c,r} \neq v_{CHN,r}]. \tag{C.2}$$

$disagreement^{I}_{c,t}$ is the share of vote disagreement between host country c and China in a given year t on issue I. $R^{I}_{c,t}$ is the set of roll calls tagged with issue I, where vote outcomes of both host and China are observed. r denotes the single roll call and $v_{c,r}, v_{CHN,r} \in \{yes, no, abstain\}$.

Tables C.5 and C.6 report 2SLS estimates using each issue as IV, alongside the baseline OLS estimate, for outcomes capturing DWF presence and fishing. The first-stage estimates show that vote disagreement on nuclear weapons and on arms control issues both predict the baseline political distance measure. For both outcomes, the 2SLS estimates are directionally similar to the OLS estimates, which indicate that the probability of the Chinese fleet being present or fishing increases as political distance widens. The 2SLS estimates based on vote

[33] All issues distinguished in Voeten (2013) are nuclear weapons and nuclear materials, arms control and disarmament, the Palestinian conflict, economic development, human rights, and colonialism.

[34] We use the simpler exact-match scoring (1 for identical votes, 0 otherwise) rather than the three-category rule of Signorino and Ritter (1999), which assigns 0.5 to abstain–yes/no pairs. Results are robust to the alternative scoring.

disagreement on nuclear weapons issue are statistically significant at the 5% level, while those based on arms control issue are less precisely estimated. The larger magnitude of the nuclear-weapons based estimates compared to the OLS results suggests the presence of a local average treatment effect, under which 2SLS identifies the effect on the subgroup of host countries whose overall political distance from China is most responsive to nuclear-issue disagreement. This is supported by our heterogeneity results in Table 3, which shows that the baseline OLS effect is concentrated among democratic hosts. These are states for whom disagreement with China on nuclear and arms-control resolutions is likely a salient marker of broader foreign-policy divergence. Together, these patterns support the interpretation that our baseline estimates capture ideology-driven political divergence rather than confounding economic ties or fisheries-specific regulatory alignment.

Table C.5: Political distance and presence 2SLS

	1(CHN fleet presence)		
	(1)	(2)	(3)
Political distance $_{t-24}$	0.151**	0.315**	0.096
	(0.066)	(0.134)	(0.129)
Estimator	OLS	2SLS	2SLS
		First stage	
		Political distance $_{t-24}$	
Vote issue		Nuclear weapons	Arms control
Vote disagreement $_{t-24}$		0.516***	1.058***
		(0.130)	(0.301)
Kleibergen-Paap F-stat		12.867	12.926
Controls	✓	✓	✓
EEZ FE	✓	✓	✓
Month-year FE	✓	✓	✓
EEZ-month FE	✓	✓	✓
Mean dep. var.	0.368	0.368	0.368
Observations	7,656	7,656	7,656

Notes: Table compares OLS (column 1) and 2SLS estimates (column 2-3) of the relationship between political distance and the probability of the Chinese presence in an EEZ buffer-month. 'CHN fleet presence' is an indicator variable equal to one if any presence hour is recorded for the Chinese fleet in the buffer, and zero otherwise. 'Political distance' is the absolute difference in UNGA voting-based ideal point between China and the EEZ's host country two years prior. The second panel reports the first stage of each 2SLS specification, where the instrument is the share of UNGA roll calls on which China and the host country voted differently on issues of Nuclear weapons and nuclear materials and Arms control (columns 2-3 respectively). Issues follow the UN issue classification distributed in `unvotes` dataset (Voeten, 2013; Robinson, 2021). Vote disagreement is constructed based on Equation C.2. All specifications include EEZ, month-year, and seasonality fixed effects, together with time-varying controls for fishing incentives (log concurrent non-Chinese fishing hours in the buffer and log phytoplankton abundance one year prior). Sample drops observations with no overlapping issue-specific votes between China and host country. This affects Comoros in 2013 on Nuclear and Arms issues, where Comoros cast no recorded votes on either issue. Standard errors clustered at the country level are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table C.6: Political distance and fishing 2SLS

	1(CHN fleet fishing)		
	(1)	(2)	(3)
Political distance $_{t-24}$	0.168*	0.326**	0.206
	(0.066)	(0.161)	(0.150)
Estimator	OLS	2SLS	2SLS
		First stage	
		Political distance $_{t-24}$	
Vote issue		Nuclear weapons	Arms control
Vote disagreement $_{t-24}$		0.516***	1.058***
		(0.130)	(0.301)
Kleibergen-Paap F-stat		12.867	12.926
Controls	✓	✓	✓
EEZ FE	✓	✓	✓
Month-year FE	✓	✓	✓
EEZ-month FE	✓	✓	✓
Mean dep. var.	0.238	0.238	0.238
Observations	7,656	7,656	7,656

Notes: Table compares OLS (column 1) and 2SLS estimates (column 2-3) of the relationship between political distance and the probability of the Chinese fleet fishing in an EEZ buffer-month. 'CHN fleet fishing' is an indicator variable equal to one if any active fishing hour is recorded for the Chinese fleet in the buffer, and zero otherwise. 'Political distance' is the absolute difference in UNGA voting-based ideal point between China and the EEZ's host country two years prior. The second panel reports the first stage of each 2SLS specification, where the instrument is the share of UNGA roll calls on which China and the host country voted differently on issues of Nuclear weapons and nuclear materials and Arms control (columns 2-3 respectively). Issues follow the UN issue classification distributed in `unvotes` dataset (Voeten, 2013; Robinson, 2021). Vote disagreement is constructed based on Equation C.2. All specifications include EEZ, month-year, and seasonality fixed effects, together with time-varying controls for fishing incentives (log concurrent non-Chinese fishing hours in the buffer and log phytoplankton abundance one year prior). Sample drops observations with no overlapping issue-specific votes between China and host country. This affects Comoros in 2013 on Nuclear and Arms issues, where Comoros cast no recorded votes on either issue. Standard errors clustered at the country level are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Appendix C.5 Robustness checks

Alternative sample selection rules

The baseline sample includes a host country only if two conditions hold. First, Chinese fishing hours in its high-seas buffers over the sample period must exceed 1,000. Second, they must be more than twice the Chinese fishing hours inside its EEZ. The buffer threshold excludes countries that are rarely reached by the Chinese fleet: their buffers are almost never visited, and the few fishing hours recorded there may stem from measurement error. The ratio rule aims to exclude countries that allow Chinese vessels to fish inside their EEZ, typically under access agreements. In these countries, Chinese presence in the buffer may simply reflect vessels moving to or from fishing grounds inside the EEZ, where they operate with the consent of the coastal state, rather than a choice to operate on the high seas next to it.

Table C.7 relaxes both rules. Columns (1) and (2) reproduce the baseline estimates of Table 1, columns 3 and 6. Lowering the ratio to one adds five countries (columns 3 and 4), lowering the buffer threshold to one hour adds 14 (columns 5 and 6), and relaxing both adds 19 (columns 7 and 8). The effect of political distance on Chinese presence ranges from 0.107 to 0.167 and remains significant at the 5 percent level in all specifications. The effect on fishing stays positive but is less precise: it is significant at the 10 percent level when only the ratio is relaxed, and the bootstrap p-values are 0.11 and 0.10 when the buffer threshold is lowered. The smaller coefficients in columns (5) to (8) are consistent with the added countries contributing mostly buffer-months without any Chinese activity.

Columns (9) and (10) take a different approach and select host countries by law rather than by observed fishing: countries whose legislation largely restricts foreign fishing in their EEZ. Foreign vessels need a permit under an international fishery agreement in the United States (United States Congress, 1976), a license in the United Kingdom and Australia (UK Parliament, 2020; Parliament of Australia, 1991), and a bilateral agreement in Iceland (Althingi, 1997). Chile and Argentina reserve their EEZs to national vessels (FAO, 1996b,a). In French waters, non-EU vessels need an EU authorization, which requires an access agreement the EU has not concluded with China (European Parliament and Council of the European Union, 2017). Ecuador has publicly opposed the Chinese fleet off the Galapagos (Stimson Center, 2020). These laws do not guarantee that no Chinese vessels are licensed, and the list of such countries is not exhaustive. In this sample of eight countries, the effect on presence is 0.105 (bootstrap p-value 0.024), while the effect on fishing falls to 0.037 and is not significant ($p = 0.113$). With only eight clusters, the bootstrap uses Webb weights.

Table C.7: Robustness: Alternative sample selection rules

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Buffer hours >	1,000		1,000		1		1		–	
Ratio >	2		1		2		1		–	
Legal restriction									✓	
	Presence	Fishing	Presence	Fishing	Presence	Fishing	Presence	Fishing	Presence	Fishing
Political distance$_{t-24}$	0.149**	0.167*	0.167**	0.139*	0.107**	0.102	0.125**	0.094	0.105**	0.037
	(0.072)	(0.091)	(0.063)	(0.076)	(0.048)	(0.061)	(0.047)	(0.057)	(0.068)	(0.034)
	[0.045]	[0.084]	[0.014]	[0.078]	[0.031]	[0.112]	[0.012]	[0.105]	[0.024]	[0.113]
Log non-CHN fishing	0.026***	0.024***	0.026***	0.026***	0.023***	0.020***	0.023***	0.021***	0.024***	0.024**
	(0.003)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)	(0.003)	(0.004)	(0.002)	(0.004)
	[0.000]	[0.001]	[0.000]	[0.000]	[0.000]	[0.001]	[0.000]	[0.000]	[0.004]	[0.011]
Log phytoplankton$_{t-12}$	0.035	0.022	0.056**	0.051*	0.019	0.013	0.037*	0.035	-0.018	-0.025
	(0.020)	(0.023)	(0.019)	(0.024)	(0.018)	(0.018)	(0.018)	(0.020)	(0.034)	(0.044)
	[0.117]	[0.372]	[0.021]	[0.072]	[0.303]	[0.480]	[0.066]	[0.116]	[0.660]	[0.602]
EEZ FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Month-year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
EEZ × month FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	7,668	7,668	8,316	8,316	9,504	9,504	10,152	10,152	6,156	6,156
Countries	21	21	26	26	35	35	40	40	8	8
EEZs	71	71	77	77	88	88	94	94	57	57

Notes: Linear probability models of Chinese fleet presence and fishing in the 50 km high-seas buffer around each EEZ, with the full set of controls of Table 1. Columns (1) to (8) vary the two sample selection rules: a country enters the sample if Chinese fishing hours in its buffers exceed the buffer threshold and are more than the stated ratio times the Chinese fishing hours inside its EEZ. Columns (1) and (2) are the baseline sample (Table 1, columns 3 and 6). Columns (9) and (10) instead include the eight countries whose legislation largely restricts foreign fishing in their EEZ. See Appendix C.5 for details.

Standard errors clustered at the country level in parentheses. Wild cluster bootstrap p-values in brackets, clustered at the country level (Rademacher weights, or Webb weights in samples with fewer than 12 countries, 9,999 replications, null hypothesis imposed). *p<0.1; **p<0.05; ***p<0.01, based on the wild cluster bootstrap p-values.

Leave-one-out tests

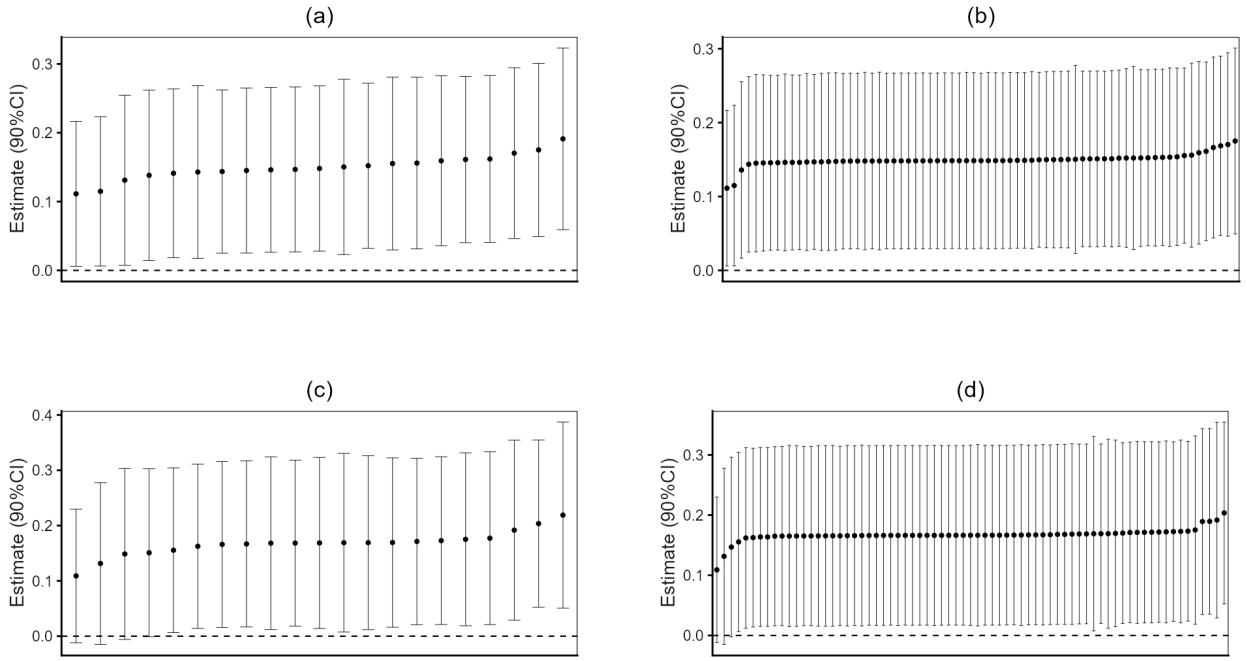


Figure C.3: Leave-one-out robustness checks

Figure plots coefficients from the main specification estimating the association between political distance from China and Chinese DWF fleet activity, with full controls as in Table 1, columns 3 and 6. Each point represents a separate regression that excludes one host country (Panels a-c) or one EEZ (Panels b-d) from the estimation sample. Outcomes are the probability of Chinese fleet presence in the buffer (Panels a-b) and the probability of active fishing (Panels c-d). Vertical bars denote 90% confidence intervals based on standard errors clustered at the country level.

Varying buffer width

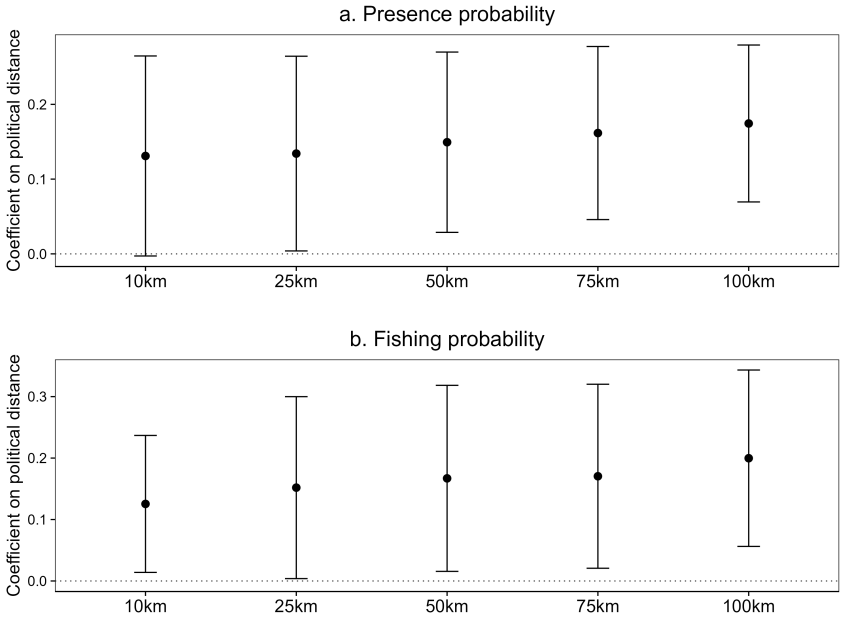


Figure C.4: Coefficients on political distance for outcomes at different buffer sizes
Figure plots coefficients on political distance obtained from the main specification (Eq. 7) regressing the probability of Chinese DWF fleet presence (Panel a) or fishing (Panel b) in the buffer just outside the host country's EEZ at various distances (x-axis). Vertical bars denote 90% confidence intervals based on standard errors clustered at the country level.

Alternative standard error clustering

Table C.8: Political distance and extensive margin of fleet presence and fishing

	1(CHN fleet presence)			1(CHN fleet fishing)		
	(1)	(2)	(3)	(4)	(5)	(6)
Political distance $_{t-24}$	-0.090**	0.177**	0.149**	-0.103***	0.191**	0.167*
	(0.035)	(0.069)	(0.066)	(0.030)	(0.086)	(0.085)
Log sea-route distance	-0.327***			-0.157**		
	(0.064)			(0.062)		
Log non-CHN fishing hours			0.026***			0.024***
			(0.005)			(0.005)
Log phytoplankton $_{t-12}$			0.035*			0.022
			(0.018)			(0.017)
EEZ FE	–	✓	✓	–	✓	✓
Month-year FE	–	✓	✓	–	✓	✓
EEZ-month FE	–	–	✓	–	–	✓
Mean dependent variable	0.368	0.368	0.368	0.238	0.238	0.238
Observations	7,668	7,668	7,668	7,668	7,668	7,668

Notes: 'CHN fleet presence' is an indicator variable equal to one if any presence hour is recorded for the Chinese fleet in the buffer, and zero otherwise. 'CHN fishing presence' is an indicator variable equal to one if any fishing hour is recorded for the Chinese fleet in the buffer, and zero otherwise. 'Political distance' is the absolute difference in UNGA voting-based ideal point between China and the EEZ's host country two years prior. 'Log sea-route distance' is the logarithmized sea-route distance from the centroid of China's EEZ to the centroid of the host EEZ. 'Log non-CHN fishing hours' is the logarithmized number of fishing hours recorded for all non-Chinese fleets in the buffer. 'Log phytoplankton' is the logarithmized amount of phytoplankton abundance measured for the buffer one year prior. Standard errors clustered at the EEZ level are reported in the parentheses. *p<0.1; **p<0.05; ***p<0.01.

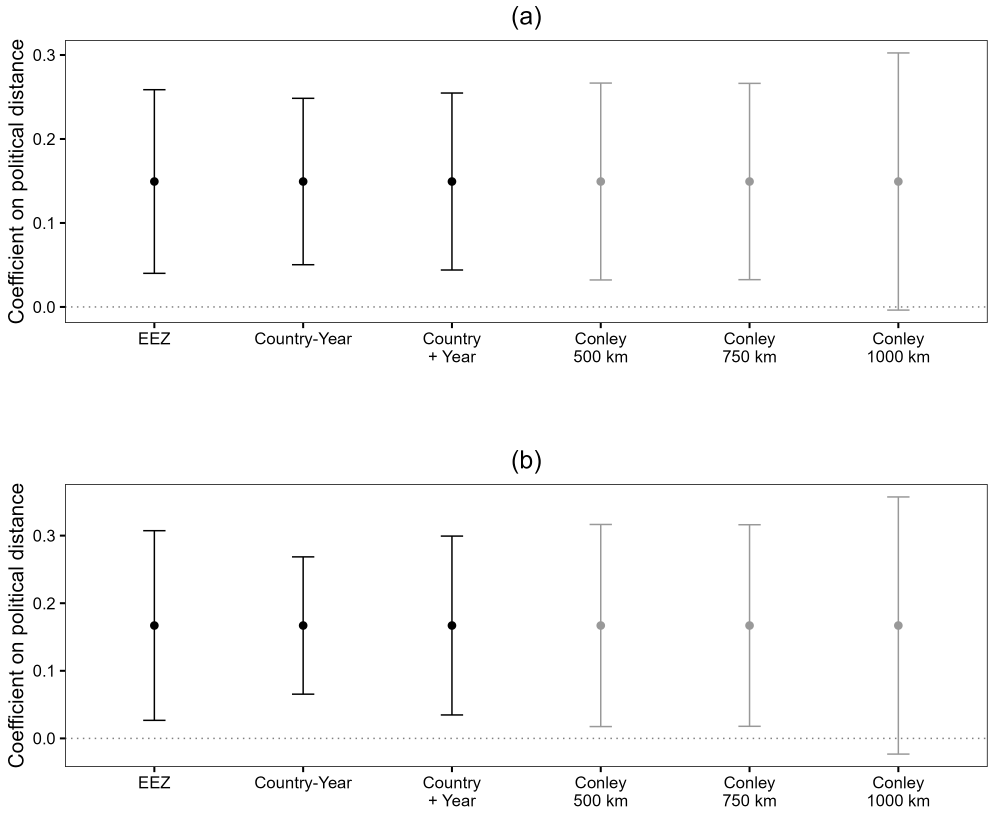


Figure C.5: Coefficients on political distance with alternative clustering

Figure plots coefficients on political distance obtained from the main specification (Eq. 7) regressing the probability of Chinese DWF fleet presence (Panel a) or fishing (Panel b) using different standard-error clustering methods (x-axis). Vertical bars denote 90% confidence intervals.

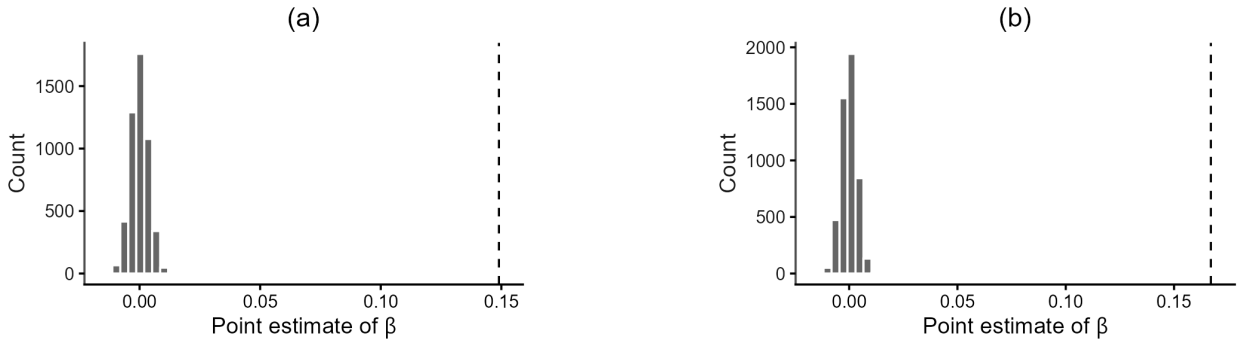


Figure C.6: Permutation inference

Figure plots the distribution of placebo coefficients on political distance under random permutation. In each iteration, we randomly reassign political distance values across host countries within a given month-year and re-estimate the main specification with full controls (Table 1, columns 3 and 6). The histogram displays the distribution of the placebo coefficients across 5,000 such permutations. The vertical line indicates the point estimate obtained from the actual data. Outcomes are the probability of Chinese fleet presence in the buffer (Panel a) and the probability of active fishing (Panel b).

Appendix D Mechanisms

Appendix D.1 Fleet-level heterogeneity

Table D.1: Heterogeneity at the fleet level: Fishing

	1(CHN fleet fishing)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Political distance $_{t-24}$	0.223**	0.207*	0.157	0.233**	0.093	-0.028	
	(0.095)	(0.099)	(0.095)	(0.098)	(0.087)	(0.155)	
	[0.018]	[0.058]	[0.129]	[0.020]	[0.361]	[0.823]	
Political distance $_{t-24}$ × Far capital	-0.401						
	(0.061)						
	[0.158]						
Political distance $_{t-24}$ × Large EEZ		-0.145					
		(0.114)					
		[0.277]					
Political distance $_{t-24}$ × Strong navy			0.026				
			(0.092)				
			[0.794]				
Political distance $_{t-24}$ × Near China				-0.278			
				(0.112)			
				[0.140]			
Political distance $_{t-24}$ × Small fleets					0.246*		
					(0.071)		
					[0.055]		
Political distance $_{t-24}$ × Democratic						0.309*	
						(0.123)	
						[0.080]	
Neg. event severity $_{t-5,\ldots,t-1}$ (std.)							0.007
							(0.009)
							[0.781]
Controls	✓	✓	✓	✓	✓	✓	✓
EEZ FE	✓	✓	✓	✓	✓	✓	✓
Month-year FE	✓	✓	✓	✓	✓	✓	✓
EEZ-month FE	✓	✓	✓	✓	✓	✓	✓
Observations	7,668	7,668	7,668	7,668	7,668	7,668	7,668

Notes: 'CHN fleet fishing' is an indicator variable equal to one if any fishing hour is recorded for the Chinese fleet in the buffer, and zero otherwise. 'Political distance' is the absolute difference in UNGA voting-based ideal point between China and the EEZ's host country two years prior. 'Far capital' equals one if the marine distance to the host country's capital exceeds the 75th percentile. 'Large EEZ' equals one if EEZ area exceeds the 75th percentile. 'Strong navy' equals one if the host country's naval tonnage exceeds the 75th percentile. 'Near China' equals one if the marine distance to China is below the 25th percentile. 'Small fleets' equals one if the average Chinese fleet size conditional on presence is below the median. Democratic' equals one if the host country's mean score on the Economist Intelligence Unit (2024) over the estimation period 2012–2020 is at least 6. 'Neg. event severity' is the cumulative sum of the Goldstein severity of all of negative events GDELT events between the host country (subject) and China (object) over the preceding five months. All specifications control for time-varying economic incentives of fishing proxied by log number of fishing hours recorded for non-Chinese fleets in the buffer and log amount of phytoplankton abundance measured for the buffer one year prior. Standard errors clustered at the country level in parentheses. Wild cluster bootstrap p-values in brackets, clustered at the country level (Rademacher weights, 9,999 replications, null hypothesis imposed). *p<0.1; **p<0.05; ***p<0.01, based on the wild cluster bootstrap p-values.

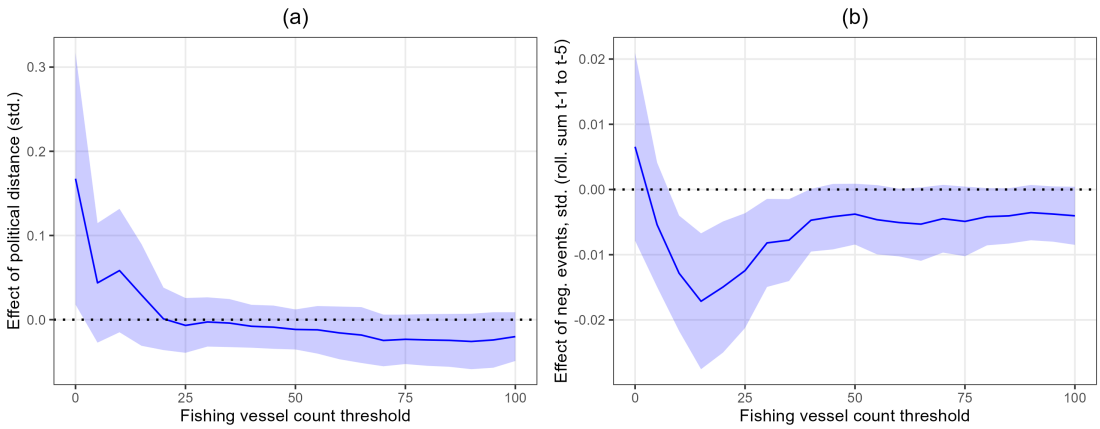


Figure D.1: Effect of political distance and negative events by fishing fleet size

Figure presents the probability of Chinese fishing with fleets above fishing-vessel-count thresholds between 0 and 100, where fishing fleet size is measured as the number of unique Chinese vessels recording positive fishing hours in the buffer that month. Every fifth threshold is estimated via an independent regression. Panel (a) shows the effect of our main political distance variable on the chance of a fishing fleet above each threshold appearing, panel (b) shows the effect of the severity-weighted rolling sum of negative events in $t-1$ until $t-5$ based on our GDELT measure. Only events in which China is the object of an action are considered in (b). Both panels show effect estimates and the corresponding 95% confidence interval and with controls similar to the main regressions.

Appendix D.2 Vessel-level heterogeneity

Table D.2 additionally show the levels of vessel characteristics of Figure 8. For vessel-level heterogeneity, we sort vessel by their own political-distance coefficient $\hat{\beta}_v$ from Eq. 8. Writing Q_q for the q-th quantile of $\hat{\beta}_v$ across the fleet, the Bottom 90% column collects vessels with $\hat{\beta}_v < Q_{0.90}$, and the Top 10%, 5% and 1% columns collect those with $\hat{\beta}_v \geq Q_{0.90}$, $Q_{0.95}$ and $Q_{0.99}$ respectively. The three Top columns are therefore nested subsets of one another, and every test reported in the table compares a Top column against the Bottom 90%, never against an adjacent column. This appendix documents the cell-level regressions that enter the table, the tests underlying the reported stars, and the definitions of all variables.

Cell-level economic response

The cell-level panel asks whether a vessel's fishing effort tracks local economic conditions at a far finer spatial resolution than the EEZ buffer. We utilize a panel of 0.1° grid cells by month covering every cell in which any Chinese vessel was ever observed, comprising 1,119,546 cells over 108 months from January 2012 to December 2020. For each vessel v and each cell-month variable $x \in \{\log NonChineseHours,\ \log Phyto_{t-12}\}$, we separately estimate

$$FishingHours_{v,g,t} = \exp(\beta_v^x x_{g,t} + \alpha_{v,g} + \alpha_{v,t})\, u_{v,g,t}, \qquad \forall\, v, \tag{D.1}$$

where g indexes grid cells, $\alpha_{v,g}$ are cell fixed effects and $\alpha_{v,t}$ month fixed effects, both estimated within vessel. The model is estimated by PPML, which accommodates the large mass of zero fishing hours without a log transformation, and β_v^x is a semi-elasticity: a unit increase in x multiplies expected fishing hours by $\exp(\beta_v^x)$.

Two features of the estimation sample deserve emphasis. First, no measure of political distance enters Eq. D.1. Political distance is a dyadic, country-level variable and has no meaning for an individual grid cell, most of which lie outside any EEZ buffer. The cell-level regressions therefore speak only to economic responsiveness, and bear on political motives solely through the contrast across columns of the table. Second, each vessel is estimated over the cells in which it recorded positive fishing hours at least once. Cells a vessel never fished contribute an outcome of zero in every month; under cell fixed effects these add nothing to the covariance between outcome and regressor but do add to the variance of the regressor, attenuating every coefficient toward zero. PPML drops such all-zero fixed-effect groups internally, so this restriction leaves the coefficients numerically unchanged.

Because the cell fixed effect absorbs all permanent differences between locations, β_v^x does *not* measure whether a vessel operates in productive regions. It measures whether, within the

cells it already works, a vessel concentrates its hours in the months when that cell is more favorable than its own average. The month fixed effect further removes variation common to all cells in a given month. Each variable enters its own regression rather than a joint specification. Coefficients are identified for 2,077 of the 2,101 vessels for $\log NonChineseHours$ and for 2,078 for $\log Phyto_{t-12}$; the remainder are vessels for which fewer than two cell-months combine positive fishing hours with a non-missing regressor.

Inference

Continuous and binary characteristics are reported as group means with standard deviations in parentheses, and stars come from a two-sample Welch t-test of each Top column against the Bottom 90%, which does not impose equal variances across the two groups.

The cell-level coefficients are treated differently. They are estimated on short vessel-specific panels and are severely heavy-tailed. That panel therefore reports group medians with the interquartile range in brackets, and stars from a Wilcoxon rank-sum test, which compares distributions through ranks and is accordingly insensitive to the tails.

Variable definitions

Vessel characteristics. 'State-owned' is the share of vessels linked to state-owned companies; 'unknown ownership' is the share of vessels for which no publicly available ownership information could be identified. 'Self-reported fishing vessel' is the share of vessels that publicly broadcast themselves as fishing vessels. 'Trawler' is the share of vessels identified as having trawler gear based on a neural network by GFW, while 'ambiguous gear' indicates cases in which the algorithm could not determine gear type from vessel behavior. 'Gear-type confidence' is a score from 0 to 1, where 1 indicates complete certainty of gear assignment by the neural network. Length and tonnage are inferred rather than registered values.

Behavioral variables. 'Log AIS disabling hours' is the logarithm of hours during which the vessel's AIS transponder is off while at sea, using the intentional disabling classification. 'Average co-present Chinese vessels' is the mean number of other Chinese-flagged vessels observed during border visits, computed as the number of distinct Chinese vessels in the same buffer-month less the vessel itself, then averaged over that vessel's visits. 'Visit-month dispersion' is the circular standard deviation of the calendar months of a vessel's border visits, computed per buffer and then averaged across buffers, where low values indicate seasonal concentration and high values indicate year-round operation. 'Share of hours in EEZ buffer' is the fraction of a vessel's total observed hours spent within the 50 km buffer immediately outside foreign EEZs.

Cell-level economic response. 'Coefficient on log non-Chinese hours' and 'coefficient on log phytoplankton$_{t-12}$' are the vessel-specific $\hat{\beta}_v^x$ from Eq. D.1: the response of a vessel's cell-level fishing hours to the log of non-Chinese vessel hours in the same cell and month and to twelve-month-lagged log phytoplankton in the same cell, respectively.

Strategic locations. All measures describe the fraction of a vessel's total observed hours spent within a stated distance of the relevant infrastructure: 50 km for submarine cables and 500 km for military installations and offshore oil platforms. Military categories follow the operator recorded in the source data, with NATO comprising United States, United Kingdom, French and Italian installations. Offshore oil proximity is matched by year, so that platforms enter only once they are observed to be flaring.

Table D.2: Vessel characteristics by political distance coefficient percentile

	Bottom 90%	Top 10%	Top 5%	Top 1%
Number of Vessels	1,890	211	106	22
Vessel characteristics				
State-owned (SOE)	0.133	0.149	0.217	0.000***
	(0.340)	(0.357)	(0.417)	(0.000)
Unknown ownership	0.622	0.521***	0.566	0.909***
	(0.485)	(0.501)	(0.498)	(0.294)
Self-reported fishing vessel	0.844	0.864	0.856	0.955**
	(0.363)	(0.344)	(0.353)	(0.213)
Trawler	0.159	0.165	0.212	0.409**
	(0.366)	(0.372)	(0.410)	(0.503)
Ambiguous gear	0.159	0.272***	0.260**	0.273
	(0.366)	(0.446)	(0.441)	(0.456)
Gear-type confidence score	0.799	0.757***	0.778	0.691***
	(0.154)	(0.145)	(0.148)	(0.135)
Behavioral patterns				
Log AIS disabling hours	2.934	2.602*	2.674	1.466***
	(3.091)	(2.729)	(2.800)	(2.339)
Average co-present Chinese vessels	86.682	59.718***	52.721***	33.086***
	(61.993)	(45.032)	(41.585)	(35.298)
Visit-month dispersion	0.750	0.985***	1.031***	1.346***
(low = seasonal, high = year-round)	(0.706)	(0.908)	(0.945)	(0.663)
Share of hours in EEZ buffer	0.174	0.257***	0.313***	0.571***
	(0.198)	(0.238)	(0.288)	(0.432)
Cell-Level Economic Response				
Coefficient on log phytoplankton$_{t-12}$	0.432	0.190***	0.099***	−0.143***
(median; IQR in brackets)	[1.047]	[0.720]	[0.612]	[0.398]
Coefficient on log non-Chinese hours	0.430	0.386	0.424	0.279
(median; IQR in brackets)	[0.808]	[0.798]	[0.788]	[0.729]
Strategic locations				
Share near submarine cable	0.318	0.512***	0.566***	0.781***
(within 50 km)	(0.252)	(0.280)	(0.286)	(0.231)
Share near NATO military base	0.179	0.218*	0.273***	0.560***
(within 500 km)	(0.294)	(0.295)	(0.339)	(0.359)
Share near non-NATO, non-Chinese base	0.006	0.014***	0.018***	0.008
(within 500 km)	(0.031)	(0.034)	(0.039)	(0.014)
Share near offshore oil platform	0.213	0.145***	0.174	0.376**
(within 500 km)	(0.300)	(0.196)	(0.230)	(0.297)

Notes: The sample consists of 2,101 Chinese DWF vessels for which a vessel-specific political-distance coefficient $\hat{\beta}_v$ is estimated via equation 8. Vessels are ranked by $\hat{\beta}_v$ and grouped into the top 10%, 5%, and 1%; the remaining bottom 90% serves as the reference group. Each cell reports the group mean (a share for binary variables) with the standard deviation in parentheses. Stars indicate the significance of a two-sample t-test of equality of means between the top group and the bottom 90%. The *cell-level economic response* panel is the exception: its entries are per-vessel PPML coefficients estimated on individual vessel-specific panels and are heavily skewed by outliers, so the panel reports group medians with the interquartile range in brackets, and stars indicate the significance of a Wilcoxon rank-sum test. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. See Appendix B.3 for details on data sources and D.2 for detailed variable definitions.

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